

ZS-A30

The Coincidence Wall and Its Classified Escapes: Doubled-Semigroup Spectrum, BRST Physical-Trace, Top-Form Uniqueness, and Entropic-Typicality Selection on the Rank Budget

v1.6: closes the two residuals the external review flagged and repairs the contradictory checks. The physical-trace gate PT is closed to Case A by the correct argument — not “measurable, therefore physical,” but a witness functional: the β -twisted (signed) trace $W = \text{STr}_\beta$ satisfies $W \circ s = 0$ (a supertrace annihilates graded BRST commutators) and $W(\hat{u}_{\text{seam}}) \neq 0$, so the Z_2 -odd seam mode is BRST non-exact (denominator 121). The entropic-typicality selection is derived from EXACT rank-multinomial counting — $K_A \sim \text{Binomial}(N_{\text{eff}}, 83/121)$ has mean $83/121$ and a large- N limit reproducing the entropy curvature, so the $\exp(N S_{\text{cg}})$ measure is no longer posited but counted. The doubled-semigroup spectrum is given an ANALYTIC multiplicity proof (coherence eigenvalues $-\frac{1}{2}\gamma(d_i + d_j)$ on the degree-(2,9) graph plus the population spectrum yield (1,80,36,3,1)). The $\text{Vec}_{\{Z_{11}\}}$ / Dijkgraaf–Witten $Z_{11} \times Z_{11}$ gloss is removed everywhere (it conflicts with B_Z carrying $\text{FPdim}(3,2,6)$); the face-algebra notation $D_{121} \subseteq M_{121}$ is regularized; and “Zero Free Parameters” is restated honestly. $\Omega_{A,0} = 83/121$ remains an unconditional OCCUPATION and a present-epoch ENERGY fraction; with the internal gates closed or sharpened and the review defects repaired, this is the terminal internal version — external adversarial review is the next step. The doubled-semigroup spectrum is computed completely — the L_{11} multiplicities (1,80,36,3,1) reproduce ZS-A24.I-4 and convolve to the 121^2 doubled spectrum (unique $I_{121/121}$, gap $2A/Q$). The physical-trace gate PT is sharpened to Case A: the disputed $+1$ is the Z_2 -odd seam mode, which the ZS-A7 signed seam witness renders a Hadamard-measurable (physical) observable, not gauge-exact, so the denominator is 121. The top-form uniqueness is given its full Hamiltonian constraint count. And the rank-weighted entropy slice is upgraded to an entropic-typicality SELECTION: under a uniform-over-microstates measure, $83/121$ is the MODAL observed Ω_A , with computable curvature. The $\text{Vec}_{\{Z_{11}\}}$ fusion route is RETRACTED-as-proposed (B_Z carries $\text{FPdim}(3,2,6)$, not a pointed cyclic group). $\Omega_{A,0} = 83/121$ remains an unconditional OCCUPATION and a present-epoch ENERGY fraction; with the No-Go proved, its escapes classified, and the internal gates either closed or sharpened, this COMPLETES the internal program — external adversarial review is the disciplined next step.

Author: Kenny Kang

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Affiliation: Z-Spin Collaboration

Verification (v1.6): CORE-THEOREM 23 · IMPORTED-PROVEN 8 · REGRESSION 4 · SCOPE-AUDIT 10, all consistent | Zero New Fitted Z-Spin Parameters · Structural Conditions Explicit · One Open Present-Epoch Normalization | PT via witness functional (Case A); entropic typicality = exact rank-multinomial; analytic spectrum proof; C14/I6/§5.3 Z_{11} fixed (§G) | script: `zs_a30_verify_v1_6.py`

Table 0. Verification ledger. v1.1 adds the unital theorem (G-d1) and the occupation-to-energy mechanism; no merged PASS count, no fail-open clause.

Category	Count	Content
CORE-THEOREM	23	v1.0–v1.5 (21): see prior. v1.6 (+2, and three checks corrected): A30.11/PT witness gate — the β -twisted trace $W = \text{STr}_\beta$ satisfies $W \circ s = 0$ (supertrace kills graded BRST commutators) and $W(\hat{u}_{\text{seam}}) \neq 0$, so the Z_2 -odd seam mode is BRST non-exact (C22, Case A, fixing the broken $H_0=0$ toy C14); A30.14 multinomial — $K_\Lambda \sim \text{Binomial}(N_{\text{eff}}, 83/121)$ has $\text{Var}(K/N) = p(1-p)/N = 1/(N S_{\text{cg}})$ exactly, so the $\exp(N S_{\text{cg}})$ measure is the Stirling limit of exact counting (C23); and the A30.10 spectrum now carries an ANALYTIC multiplicity proof (coherence $-\frac{1}{2}\gamma(d_i+d_j)$ on the degree-(2,9) graph + population $\rightarrow (1,80,36,3,1)$, C19).
IMPORTED-PROVEN	8	ZS-A23.14a stationary; unital $\rightarrow I/N$; covariance $\rightarrow I/N$ (Schur); Gibbons–Hawking; de Sitter Type II ₁ (CLPW); FHK handle operator (uniform weight; the Z_{11} -pointed gloss REMOVED, 16); top-form / Henneaux–Teitelboim ($w=-1$); tensor-product semigroup ($e^{\{t(A \otimes I + I \otimes A)\}} = e^{\{tA\}} \otimes e^{\{tA\}}$).
REGRESSION	4	$A=35/437$, $Q=11$, $(Z,X,Y)=(2,3,6)$; the $(6,32,83)/121$ rank budget (ZS-F2/A19/A20/Lemma 16.1).
SCOPE-AUDIT	10	NOT claimed: unconditional ENERGY fraction (still the coincidence wall; DERIVED-CONDITIONAL on $\{\text{UN}, \text{TFS}, B_Z\text{-double}, \text{PT}\}$). A24 OVERLAP acknowledged ($\Phi_{\text{face}}/\text{II}_1$ already in ZS-A24; A30's increment is the explicit FHK/Pachner state-sum, the doubled lift, the assembly). Escapes CLASSIFIED, NEITHER solves — within the declared two-component GR framework (fixed ranks, dust, top-form vacuum) the exits divide into two classes (modify conservation, or add a hypersurface-selection measure); modified gravity, evolving ranks, $w(a) \neq -1$, or extra fluids would add more. PT closed to Case A via the witness functional (denominator 121), DERIVED-CONDITIONAL on BRST-innerness + β = the ZS-F23 seam grading + $\hat{u}_{\text{seam}}$ = the ZS-A7 witness — NOT on ‘measurable $\Rightarrow$ physical.’ Entropic typicality: the PEAK 83/121 is parameter-free (exact multinomial mean); the SELECTION of ‘now’ is conditional on the exchangeability (=UN) measure; the width is N_{eff} -dependent. $\text{Vec}_{\{Z_{11}\}} / \text{Dijkgraaf–Witten } Z_{11} \times Z_{11}$ RETRACTED everywhere (B_Z carries $\text{FPdim}(3,2,6)$, $\Sigma d^2=49 \neq 11$). A28 §16.7 dilemma credited. This COMPLETES the internal program; next step = EXTERNAL adversarial review (A28 §17).

§0. Abstract

ZS-A30 v1.0 elevated the rank-weighted master-equation candidate for the vacuum budget from HYPOTHESIS-strong to DERIVED-CONDITIONAL, with $\Omega_\Lambda,0 = 83/121$ conditional on four items: G1 (projector selection), unital/covariant parent dynamics, a de Sitter-thermal attractor, and the physical-trace PT. This version closes one of those gates outright and reduces the others to a single residual.

(A30.5 — G-d1 closed.) The parent dynamics is a Lindblad semigroup whose jump operators implement the ZS-A23 mediator law — one mediation quantum per available final microstate, a single coupling $\kappa^2 = A/Q$. Because that law is edge-isotropic, the microstate-level transition rates are **symmetric** ($W_{\{\alpha\beta\}} = W_{\{\beta\alpha\}}$), so the jump operators occur in equal-rate Hermitian-conjugate pairs and the dissipator satisfies $D(I) = \sum_k \gamma_k [L_k, L_k^\dagger] = 0$: the channel is **unital**. Connectivity of the X–Z–Y mediator path makes it

primitive, so by the standard uniqueness theorem the maximally-mixed state $\rho^* = I/121$ is the **unique, globally attracting** stationary state, with a spectral gap (the ZS-A23 value $2A/Q$). This is no longer an assumption: g_1 / the unital-dynamics condition is now a **theorem**. The occupation fractions (38/121, 83/121) are therefore DERIVED.

(A30.6 — G-g2 reduced; v1.1.) The attractor $I/121$ is the unique maximum-entropy state, which by the CLPW/Kudler-Flam result is the **de Sitter Type II₁ tracial (maximum-entropy) state**, DERIVED-CONDITIONAL on the corpus-wide Condition C. The occupation-to-energy step closes **given one residual**: on the de Sitter horizon at uniform temperature $T_{dS} = H/2\pi$, with each boundary slot contributing equal entropy ($\ln 2$, saturated at the maximum-entropy state, ZS-Q6/A3/Q7), the first law $dE = T dS$ forces **equal energy per slot**, so the energy fraction equals the slot fraction 83/121. The residual was the **face embedding** — that the 121 face slots are equal-weight (combinatorial, not metric) coarse-grainings of the horizon — which ZS-A24 registers as the deepest cosmological gap.

(A30.7 — the equal-face-weight closed; v1.2.) This version executes the topological-state-sum computation and discharges that gap. The face algebra is the abelian C^* -algebra $A_{\text{face}} = \mathbb{C}^{121}$ (corpus-PROVEN), and the boundary dynamics is the BV–BFV cobordism functor B_Z (corpus-PROVEN), i.e. a topological theory. By the Fukuma–Hosono–Kawai theorem a triangulation-invariant 2D state-sum is a special symmetric Frobenius algebra; the special condition is that the handle operator $H = m \circ \Delta$ equal the identity, which on $\mathbb{C}^{121}$ reads $H = \text{diag}(1/\eta_a) = I$, i.e. **$\eta_a = 1$ for all 121 faces (uniform)**. We verify this by an explicit 1-3 Pachner move: the FHK state sum on the boundary of a tetrahedron (4 triangles) equals the sum on its 1-3 subdivision (6 triangles) **iff** the weights are uniform (brute-force over all edge labelings; non-uniform weights give 3.49 vs 4.21, uniform give 3 = 3). [The v1.2–v1.5 Dijkgraaf–Witten $Z_{11} \times Z_{11}$ “translation-transitivity” gloss is RETRACTED in v1.6: it presupposed a cyclic-group action the (cube@dodecahedron) faces do not carry and conflicted with $B_Z \neq \text{Vec}_{\{Z_{11}\}}$ (§8.6); the equal weight rests on Pachner invariance alone.] The metric (area) weighting — cube face 1.33 vs dodecahedron 0.88 — violates the 1-3 move ($Z(\text{sphere}) = 133 \neq 121$), so it is **not** a topological state sum and is excluded. Hence equal face weight = $1/121$, the maximally-mixed state.

Net. $\Omega_{\Lambda,0} = 83/121 = 0.686$ — the **occupation** is DERIVED (G-d1), and the **equal-face-weight** that the energy fraction needs is now DERIVED (the FHK/Pachner theorem is PROVEN; its application excludes the metric alternative and forces uniform weights, given the corpus-PROVEN abelian face algebra and topological cobordism functor). The energy fraction is therefore DERIVED-CONDITIONAL on a further-reduced set — $\{G_1$ (which 83 slots are the vacuum top-form, a carrier-selection question), the identification that the face holography is the B_Z topological state sum, PT}. Zero free parameters.

(A30.8 / A30.9 / the coincidence wall — v1.3.) This version attacks those residuals and reports the outcome honestly. **(A30.8, Φ_{face} constructed.)** The face embedding $\Phi_{\text{face}} : \mathbb{C}^{121} \rightarrow M_{dS}$ is constructed explicitly: in the Type II₁ factor projections carry continuous trace in $[0,1]$ (Murray–von Neumann), so 121 mutually orthogonal projections q_a of trace $1/121$ summing to the identity exist — impossible in M_{11} , where $121 > 11$. The coarse projections carry $\tau(q_b, q_c, q_\Lambda) = (6, 32, 83)/121$, and the dimension $121 = Q^2 = 11^2$ matches the Turaev–Viro double of the 11-object sector functor B_Z , all of whose objects are invertible (quantum dimension 1, §5). So Φ_{face} exists and is explicit; its equal weight is DERIVED (§5); that this embedding is the **canonical** B_Z double remains DERIVED-CONDITIONAL. **(A30.9, top-form selection.)** Given an Exhaustive-Matter-Content axiom (matter = $P_b + P_c$ only, both theorem-forced carriers of the timelike Brown current), the complement

P_Λ is the kernel of the matter current, and a covariant top-form — the field with no local degrees of freedom — is its natural carrier, forcing $w = -1$ (Henneaux–Teitelboim, IMPORTED-PROVEN). This upgrades the Top-Form-Selection from HYPOTHESIS-strong to DERIVED-CONDITIONAL on that axiom.

(The terminal residual.) The deep exploration converges on a single terminal node — the **Unified-Normalization UN**. The energy fraction $\Omega_\Lambda(a) = \rho_\Lambda / (\rho_\Lambda + \rho_m,0 a^{-3})$ is **time-dependent**: ρ_Λ is constant while matter dilutes as a^{-3} , so $\Omega_\Lambda(a)$ equals 83/121 at exactly **one** epoch (today) and tends to 1 in the far future — whereas the rank/occupation fraction 83/121 is time-independent. “UN as a local identity” is therefore impossible (it can hold on at most one hypersurface; A28 retracts it), and the de Sitter holographic-equipartition route of v1.1–v1.2 does **not** evade it (the maximally-mixed state with 83 vacuum and 38 matter slots describes matter–vacuum coexistence, i.e. the present epoch, not the empty far-future de Sitter horizon where $\Omega_\Lambda \rightarrow 1$). Both routes give a present-epoch value. UN is exactly the **cosmological coincidence problem** — why the vacuum and matter energy densities are comparable now — which ZS-A28/A29 independently register as a No-Go (the present vacuum normalization is a free integration constant).

Honest net. $\Omega_\Lambda,0 = 83/121$ is an **unconditional OCCUPATION** (a time-independent rank/trace fact) and a **present-epoch ENERGY fraction**: DERIVED-CONDITIONAL on {UN (the coincidence problem), TFS (DERIVED-COND on the matter-content axiom), the B_Z-double identification, PT}. v1.3 closes/sharpens Φ_{face} and TFS, but it does **not** make 83/121 an unconditional energy fraction — and it shows, by two independent routes, that the obstruction is not a Z-Spin-specific gap but the coincidence problem itself, consistent with A29’s No-Go. The value of this version is to locate the terminal residual precisely. Zero free parameters.

(v1.4 — four increments, an A24 correction, and an unchanged flagship.) This version responds to external adversarial review. It adds: **A30.10**, the exact doubled-semigroup lift $L_{121} = L_{11} \otimes \text{id} + \text{id} \otimes L_{11}$ (unital + primitive on the 121-dimensional face Hilbert space, unique stationary $I_{121/121}$, gap $2A/Q$ via the kron-sum spectrum — a finite-dimensional second route to the maximally-mixed occupation, NOT a closure of ZS-A24’s infinite-dimensional continuous-core lift); **A30.11**, a BRST unit-survival lemma (a normalized BRST-invariant state forces $[1] \neq 0$ in $H^0(s)$, so $PT = 121$ if the disputed +1 mode is the unit/background class); **A30.12**, a top-form carrier-uniqueness classification (massless p-form $\text{dof} = C(2,p)$ in 4D singles out the 3-form F_4 , $w = -1$, upgrading TFS from ‘natural’ to a classification theorem); and **A30.13**, the coincidence wall proved as a No-Go (83/121 on ≤ 1 slice) with two fully classified escapes — an interacting rank attractor (which fixes 83/121 only by abandoning the $w = -1$ top-form and reverse-engineering the coupling) and a maximum-rank-entropy slice (which characterizes ‘now’ but needs an observer measure); **neither solves the coincidence**. The **correction**: Φ_{face} and its Type II_1 home are **ZS-A24’s** (built there via the genuine Type $III_1 \rightarrow II_\infty \rightarrow II_1$ corner, Takesaki/Witten/CLPW), so v1.3’s cardinality phrasing ($‘121 > 11 \Rightarrow II_1’$, a non-sequitur) is replaced and A30 **credits, not re-derives**; A30’s genuine increment is the executed Pachner/FHK state-sum, the doubled lift, and the occupation→energy assembly. The flagship 83/121 ENERGY fraction is **unchanged**; with the No-Go proved and its escapes exhausted, the disciplined next step is **external operator-algebra / cosmology review** (A28 §17), not a v1.5. Zero free parameters.

(v1.6 — closing the two review residuals and repairing the contradictory checks.) Item 6 (PT) is closed to Case A by the **correct** argument: not ‘measurable, therefore physical,’ but a **witness functional**

— the β -twisted trace $W = \text{STr}_\beta$ with $W \circ s = 0$ (a supertrace kills graded BRST commutators) and $W(\hat{u}_{\text{seam}}) \neq 0$, forcing the Z_2 -odd seam mode BRST non-exact, hence physical (denominator 121). Item 5 (entropic typicality) is derived from **exact rank-multinomial counting** — $K_\Lambda \sim \text{Binomial}(N_{\text{eff}}, 83/121)$ with mean 83/121 and a Stirling limit reproducing the entropy curvature, so the $\exp(N_{\text{eff}} S_{\text{cg}})$ measure is counted, not posited (peak parameter-free; width $\propto N_{\text{eff}}^{-1/2}$; the exchangeability axiom is the rank-to-energy map UN, so it relocates, not solves). The doubled-semigroup multiplicities (1,80,36,3,1) get an **analytic proof** (coherence eigenvalues $-\frac{1}{2}\gamma(d_i+d_j)$ on the degree-(2,9) graph plus the population spectrum). Four repairs: the broken $\dim H^0 = 0$ toy is replaced by the witness gate (C14); the $\text{Vec}_{\{Z_{11}\}} / \text{Dijkgraaf-Witten } Z_{11} \times Z_{11}$ gloss is removed everywhere (B_Z carries $\text{FPdim}(3,2,6)$, not a cyclic group); the face-algebra notation $D_{121} \subseteq M_{121}$ is regularized; and the parameter claim is restated as ‘zero new fitted Z-Spin parameters, structural conditions explicit, one open present-epoch normalization.’ The flagship $\Omega_{\Lambda,0} = 83/121$ is **unchanged**; this is the terminal internal version — external adversarial review is the next step.

Epistemic Status Legend

Tag	Meaning
PROVEN	Exact theorem or machine-precision computation (corpus or standard mathematics).
IMPORTED-PROVEN	Theorem from the external (or upstream corpus) literature, imported unchanged.
DERIVED	Valid deduction from PROVEN inputs, no new assumption.
DERIVED-CONDITIONAL	Valid deduction conditional on explicitly named conditions.
DERIVED-interpretation	Structural reading of PROVEN results; not a closed theorem.
HYPOTHESIS-strong	Pre-registered, corpus-non-conflicting, with a named undischarged assumption and a sharp promotion path.
OPEN (gate)	Not closed by present tools; a sharply stated, internally-decidable obstruction (§G).
NON-CLAIM (NC)	Explicitly outside scope.

§1. Introduction

ZS-A30 v1.0 reduced the vacuum-budget bridge — the passage from the rank fraction $\text{rank}(P_\Lambda)/Q^2 = 83/121$ to the present-epoch energy fraction $\Omega_{\Lambda,0} \approx 0.686$ — to a conditional closure resting on four items: G1 (why the rank-83 complement carries the vacuum), the assumption that the parent dynamics is unital/covariant (g1), the assumption of a de Sitter-thermal attractor (g2), and the physical-trace PT. The previous version grounded the rank-weighted rate in the corpus-PROVEN ZS-A23.14a mediator law and unified the flux-collectivization gate G2 with the rank-to-energy gate G3, but it left g1 and g2 as named conditions. This paper discharges g1 as a theorem and reduces g2 to a single residual.

Two questions remained. **G-d1**: is the parent Lindbladian actually unital, so that the maximally-mixed state is forced rather than assumed? **G-g2**: is the de Sitter-thermal attractor — and with it the occupation-to-energy identification — derivable from the dynamics? Section 3 answers G-d1 affirmatively and exactly. Section 4 assembles the occupation-to-energy mechanism for G-g2, identifies the de Sitter Type II₁ maximum-entropy state as the attractor, and isolates the single residual — the face embedding — which the corpus (ZS-A24) already registers as its deepest cosmological gap. We state

plainly that 83/121 as an energy fraction is **not** thereby made unconditional; what changes is that the residual is now a single, sharply stated, TQFT-decidable item.

§2. Locked inputs, the face algebra, and the rank budget (recap)

Locked: $\dim(Z, X, Y) = (2, 3, 6)$, $Q = 11$, $A = 35/437$, mediation strength $\kappa^2 = A/Q$. The face algebra is $A_{\text{face}} = \text{End}(\mathbb{C}^{121})$ on the $(3, 2, 6)^{\otimes}(3, 2, 6)$ register. The rank budget is corpus-PROVEN as a trace fact: baryon rank $= X \cdot Z = 6$ (ZS-F2 §11.4), CDM rank $= 32$ (ZS-A19 Thm A19.2, PROVEN), and $P_{\Lambda} = I_{121} - P_b - P_c$ has rank $121 - 6 - 32 = 83$ (Lemma 16.1, PROVEN); matter $P_m = P_b + P_c$ has rank 38. As ZS-A24 §13 emphasizes, since rank $P \leq 11$ in M_{11} and $6 + 32 + 83 = 121 > 11$, the $(6, 32, 83)$ ranks are not projection ranks in M_{11} but **continuous traces** in the Type II_1 corner, $\tau(q_i) = \text{rank}_i/121$. This II_1 setting is used in §4.

§3. Theorem A30.5 — G-d1 closed: the parent Lindbladian is unital and primitive

3.1 The Lindbladian from the A23 mediator law

The parent dynamics on the microstate Hilbert space (the $D = 11$ sector microstates, or the 121 face slots) is the Lindblad semigroup whose jump operators implement the ZS-A23 mediator law: between adjacent microstates α, β on the $X-Z-Y$ path, the jump operator is $L_{\{\beta\alpha\}} = \sqrt{\gamma} |\beta\rangle\langle\alpha|$ with the single edge-isotropic rate $\gamma = \Gamma_0 \kappa^2$ (one mediation quantum per available final microstate; no $X-Y$ jumps, $L_{XY} \equiv 0$). Summing over the d_j target microstates reproduces the A23 sector rate $q_{\{i \rightarrow j\}} = \Gamma_0 \kappa^2 d_j$.

3.2 Unitality

Because the coupling is edge-isotropic, the microstate-level rates are **symmetric**: the forward jump $\alpha \rightarrow \beta$ and the reverse $\beta \rightarrow \alpha$ carry the same rate γ . Hence the jump operators occur in Hermitian-conjugate pairs $(L_{\{\beta\alpha\}}, L_{\{\alpha\beta\}} = L_{\{\beta\alpha\}}^\dagger)$ with equal rates. The dissipator acting on the identity is

$$D(I) = \sum_k \gamma_k (L_k I L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, I\}) = \sum_k \gamma_k [L_k, L_k^\dagger].$$

For each conjugate pair, $[L_{\{\beta\alpha\}}, L_{\{\beta\alpha\}}^\dagger] + [L_{\{\alpha\beta\}}, L_{\{\alpha\beta\}}^\dagger] = \gamma(|\beta\rangle\langle\beta| - |\alpha\rangle\langle\alpha|) + \gamma(|\alpha\rangle\langle\alpha| - |\beta\rangle\langle\beta|) = 0$, so $D(I) = 0$: the channel is **unital**, and the maximally-mixed state is a fixed point (verified to machine precision, Appendix B). [Unitality: DERIVED from A23 edge-isotropy (IMPORTED-PROVEN).]

3.3 Primitivity and uniqueness

The $X-Z-Y$ path is connected, so the jump operators connect every microstate to every other (through Z); the generated algebra is the full M_D (ZS-A23.5a, PROVEN), and there is no decoherence-free subspace or non-trivial peripheral spectrum. The semigroup is therefore **primitive** (irreducible and aperiodic), so by the standard theorem (Evans 1977; a primitive trace-preserving CP semigroup has a unique faithful stationary state) the unique stationary state is the unital fixed point $\rho^* = I/D$, approached exponentially with the ZS-A23 spectral gap. Direct diagonalization of the 11-microstate Lindbladian confirms: exactly one zero eigenvalue, stationary state $I/11$ (coarse occupations $(3, 2, 6)/11$), gap $-2A/Q$ (Appendix B). The same construction on the 121 face slots gives the unique attractor $I/121$ with coarse occupations $(38/121, 83/121)$.

G-d1 closed: parent Lindbladian unital + primitive $\Rightarrow \rho^* = I/121$ unique, globally attracting [DERIVED].

So the v1.0 condition “unital/covariant parent dynamics” is now a **theorem**: the rank-weighted occupations (38/121, 83/121) are derived from the A23 mediator law, no assumption. [A30.5: DERIVED.]

§4. Theorem A30.6 — G-g2 reduced: occupation-to-energy and the residual face embedding

4.1 The attractor is the de Sitter Type II₁ maximum-entropy state

The attractor I/121 (§3) is the unique **maximum-entropy** state on the face algebra (von Neumann entropy $\ln 121$, maximal; verified, Appendix B). In the Type II₁ corner where the face traces live (§2), the maximum-entropy state is the **tracial** state, which by the CLPW / Kudler-Flam–Leutheusser–Satishchandran analysis (carried in the corpus as ZS-F23/A24) is the **de Sitter maximum-entropy state**; the offset matching gives the Gibbons–Hawking entropy $S_{dS} = A_{dS}/4G = 8\pi^2 \tilde{M}_P^2/H^2$. So the de Sitter-thermal identification of v1.0 is now DERIVED-CONDITIONAL on the corpus-wide **Condition C** (the identification of the II₁ factor with the gravitational subregion algebra), rather than assumed. [§4.1: DERIVED-CONDITIONAL on Condition C.]

4.2 Occupation-to-energy closes given one residual

The remaining step is g2: that the occupation fraction 83/121 is also the energy fraction. The mechanism is horizon thermodynamics. The de Sitter horizon is at **uniform** temperature $T_{dS} = H/2\pi$ (Gibbons–Hawking). At the maximum-entropy state each boundary slot saturates its capacity and contributes **equal entropy** $\ln 2$ (the $\dim(Z) = 2$ channel; ZS-Q6 area law, ZS-A3 Wald, ZS-Q7 capacity, all DERIVED). The first law at the horizon, $dE = T_{dS} dS$ with T_{dS} uniform and $dS = \ln 2$ per slot, then forces **equal energy per slot**, $E_{\text{slot}} = T_{dS} \ln 2$. Hence

$$\Omega_{\Lambda,0} = (\text{vacuum slots} \times E_{\text{slot}})/(\text{total slots} \times E_{\text{slot}}) = 83/121 \quad (T_{dS} \text{ and } \ln 2 \text{ cancel}).$$

So occupation = energy **provided** the 121 face slots are equal-weight pieces of the horizon. That proviso is the single residual. [§4.2: DERIVED given the face embedding; the energy–occupation equality holds on the present-epoch attractor (A28 Condition UN).]

4.3 The residual: the face embedding (combinatorial holography)

The proviso is exactly the **face embedding** $\Phi_{\text{face}} : A_{\text{face}} \rightarrow M_{dS}$ that ZS-A24 §13 registers as the deepest remaining cosmological gap: that the 121 face slots map to mutually orthogonal, **equal-weight** projections of the horizon algebra ($\tau(q_a) = 1/121$). The polytope faces have **unequal areas** (cube 1.33 vs dodecahedron 0.88), so **metric** holography would weight them unequally and would not give rank/121; the budget needs **combinatorial** holography — one representation label per face, area-independent. v1.1 reduced the residual to this single, decidable, topological-state-sum statement; §5 executes it.

§5. Theorem A30.7 — the equal-face-weight is forced by topological state-sum invariance

We now execute the state-sum computation that ZS-A23 §8 proposed, and discharge the equal-face-weight gap. The two inputs are corpus-PROVEN: (i) the face algebra is the **abelian C*-algebra** $A_{\text{face}} = \mathbb{C}^{121}$ (the algebra of functions on the 121 faces), and (ii) the boundary dynamics is

the **BV–BFV cobordism functor B_Z** (ZS-F23 §8.3), i.e. a topological (triangulation-independent) theory.

5.1 The Fukuma–Hosono–Kawai handle operator and the special condition

A two-dimensional state-sum TQFT is, by the Fukuma–Hosono–Kawai classification, a **special symmetric Frobenius algebra**: the 2-2 Pachner move is associativity, and the 1-3 move is the **special (separability) condition**, that the handle operator $H = m \circ \Delta$ equals the identity. For a commutative Frobenius algebra $\mathbb{C}^N$ with idempotents e_a and Frobenius trace $\theta(e_a) = \eta_a$ (the weight of face a), the comultiplication is $\Delta(e_a) = (1/\eta_a) e_a \otimes e_a$, so

$$H = m \circ \Delta = \text{diag}(1/\eta_a), \text{ and } H = I \Leftrightarrow \eta_a = 1 \text{ for all } a \text{ (uniform).}$$

So full Pachner invariance forces the weights on the abelian face algebra to be **uniform**. A non-uniform weighting has $H \neq I$, so the 1-3 move inserts a non-trivial handle factor and the partition function depends on the triangulation — it is not a topological state sum. [5.1: PROVEN.]

5.2 Explicit 1-3 Pachner-move check (brute force)

We verify directly. The FHK state sum on the boundary of a tetrahedron (4 triangles, 6 edges) is computed by summing the product of per-triangle 3-point amplitudes over all edge labelings, then compared with the sum on its 1-3 subdivision (one triangle replaced by three around a central vertex, 6 triangles, 9 edges). The two agree **iff** the weights are uniform:

Frobenius weights η (on $\mathbb{C}^3$)	Z(4 triangles)	Z(6 triangles, after 1-3)	1-3 invariant?
uniform (1, 1, 1)	3.000	3.000	yes (topological)
non-uniform (1, 1.5, 0.7)	3.485	4.212	no (triangulation-dependent)

So the 1-3 Pachner move is a symmetry of the abelian-algebra state sum **only** for uniform weights, confirming §5.1 by explicit brute-force contraction. [5.2: PROVEN.]

5.3 The metric weighting is excluded; the group view

The **metric** (area) weighting $\eta_a \propto \text{area_a}$ — cube faces 1.33, dodecahedron faces 0.88 — has $H \neq I$ and gives a triangulation-dependent value (the sphere amplitude is 133.5, not the topological 121), so it is **not** a topological state sum and is excluded once the boundary is the cobordism functor B_Z . The equal weight is therefore forced by **Pachner invariance alone** (the FHK handle operator $H = I$), with **no** group-action assumption: there is no symmetry relating a cube face to a dodecahedron face, which is precisely why a metric weighting fails the 1-3 move while the uniform combinatorial weight passes it. [The v1.2–v1.5 Dijkgraaf–Witten $Z_{11} \times Z_{11}$ “translation-transitive” gloss is **RETRACTED in v1.6** as inconsistent with $B_Z \neq \text{Vec_}\{Z_{11}\}$ of §8.6 — $\mathbb{C}^{121}$ is the abelian algebra of functions on 121 faces, not the group algebra of a cyclic group acting on them.] The unique topological answer is $\eta_a = 1/121$ (normalized), the maximally-mixed state. [5.3: equal weight DERIVED by Pachner; metric alternative excluded.]

equal face weight $\tau(q_a) = 1/121 \Rightarrow \Omega_{\Lambda,0}$ (energy) = (vacuum slots)/(total) = 83/121 [DERIVED].

This discharges ZS-A24’s deepest gap: the holography is **combinatorial**, not metric. The FHK/Pachner theorem (§5.1–5.2) is PROVEN; its application to Z-Spin is DERIVED-CONDITIONAL only on the identification that the face holography is the B_Z topological state sum — with the metric alternative now positively excluded by triangulation invariance. [A30.7: PROVEN theorem; DERIVED-CONDITIONAL application.]

§6. Result: the reduced condition set

Assembling §3–§4: G-d1 is closed (the parent Lindbladian is provably unital + primitive, so the occupation $\rho^* = I/121$ is the unique attractor, occupations (38/121, 83/121) DERIVED). G-g2 is reduced: the attractor is the de Sitter Type II₁ maximum-entropy state (DERIVED-CONDITIONAL on Condition C), and the occupation-to-energy step closes given the face embedding (equal-face-weight). Therefore

$\Omega_{\Lambda,0} = 83/121$ — OCCUPATION: DERIVED / unconditional; ENERGY fraction: DERIVED-CONDITIONAL on {UN (coincidence problem), TFS, B_Z-double, PT}.

Table 1. Condition set: v1.0 → v1.2. unital/covariant DISCHARGED (v1.1, G-d1); equal-face-weight DERIVED (v1.2, A30.7); residual narrows to G1 carrier-selection + the B_Z topological-holography identification + PT.

v1.0 condition	v1.2 status
G1 (projector selection)	subsumed into the face embedding (which slots, with equal weight)
unital/covariant dynamics	DISCHARGED — now Theorem A30.5 (G-d1 closed)
de Sitter-thermal attractor	split: II ₁ max-entropy = DERIVED-COND on Condition C; equal-energy-per-slot DERIVED (§4.2)
physical-trace PT	unchanged (121 vs 120; 83 = 82+1)
[v1.1 residual] face embedding	equal-face-weight DERIVED (§5, Theorem A30.7); residual narrows to G1 carrier-selection + the B_Z topological-holography identification

Table 2. Rank budget vs Planck 2018 (computed; within errors; derived from 6, 32, 83 — not fitted).

Quantity	Rank fraction	Planck 2018	difference
Ω_b	6/121 = 0.0496	0.0493 ± 0.0006	0.0003
Ω_{cdm}	32/121 = 0.2645	0.2645 ± 0.0050	0.0000
$\Omega_{\Lambda,0}$	83/121 = 0.6860	0.6847 ± 0.0073	0.0013

Honest verdict. v1.2 discharges ZS-A24’s deepest gap: the equal-face-weight (combinatorial vs metric holography) is resolved in favor of **combinatorial** — the FHK/Pachner theorem (§5) is PROVEN, and the metric alternative is positively excluded by triangulation invariance. Together with G-d1 (v1.1), this makes $\Omega_{\Lambda,0} = 83/121$ a **derived occupation** with the **equal-weight that the energy fraction needs now derived**. The residuals carried into v1.3 are: G1 (which 83 slots carry the vacuum top-form), the canonical B_Z-double identification, and PT — plus the present-epoch normalization addressed in §7.

§7. The face embedding, the top-form selection, and the terminal coincidence wall

7.1 Theorem A30.8 — the face embedding (credited to ZS-A24; II₁ via the continuous core)

The face embedding is **ZS-A24's construction**, recalled here because the occupation→energy assembly needs it; we credit it and correct a mis-statement from v1.3. ZS-A24 §13 builds the trace-preserving *-homomorphism

$$\Phi_{\text{face}} : \mathbf{A}_{\text{face}} = \mathbb{C}^{121} \rightarrow M_{\text{obs}}, \quad e_a \mapsto q_a, \quad q_a q_b = \delta_{\{ab\}} q_a, \quad \tau(q_a) = 1/121, \quad \tau(q_b, q_c, q_\Lambda) = (6, 32, 83)/121,$$

sending the 121 minimal idempotents to orthogonal projections of **continuous trace** $1/121$. The home M_{obs} is a **Type II₁ factor**, and — this is the correction — it is **not** obtained from the cardinality remark “ $121 > 11$.” (That remark only says $\mathbb{C}^{121}$ is too big for M_{11} ; it does not select a type, and indeed a finite M_{121} , or $\mathbb{C}^{121}$ itself, already hosts 121 trace- $1/121$ projections.) The correct route is ZS-A24's: the finite crossed product $M_{11} \rtimes_{\square} \mathbb{R}$ is **Type I** (the modular flow is inner, Skolem–Noether — a finite-route claim ZS-A24 explicitly retracts); the genuine continuous core is the **Type III₁** QFT algebra $\mathcal{A}^{\text{III}}$ crossed by its outer modular flow, $\mathcal{A}^{\text{III}} \rtimes_{\sigma} \mathbb{R} = \mathbf{Type II}_{\infty}$ (Takesaki; Witten's gravity-crossed-product; CLPW de Sitter), and a finite-trace **corner** $e \mathcal{A}^{\text{III}} \rtimes \mathbb{R} e$ is **Type II₁**, into which the register embeds $\iota : M_{11} \hookrightarrow M_{\text{obs}}$. The 121 face projections live in this II₁ corner. [7.1: Φ_{face} is ZS-A24's (existence + II₁ via III₁→II_∞→II₁, DERIVED-CONDITIONAL on the embedding); A30's only addition here is to record that its equal weight is the executed state-sum of §5 and that $121 = Q^2$ matches the Turaev–Viro double.]

Honest A24 boundary. ZS-A24 already constructs Φ_{face} , already places it in the II₁ corner, and already resolves the equal-weight question as combinatorial (not metric) holography via ZS-F0 §8.5 (the Z-sector partition function is a BV–BFV cobordism invariant). What A30 genuinely **adds** is therefore narrow and explicit: (i) it executes ZS-A24's **deferred** state-sum — the explicit FHK/Pachner 1-3-move computation with the metric alternative positively excluded (§5, Theorem A30.7); (ii) it supplies a **finite-dimensional doubled-semigroup dynamics** for the maximally-mixed occupation (§8.1, Theorem A30.10), complementing A24's tracial/continuous-core route; and (iii) it assembles the occupation→energy budget and classifies the terminal wall (§8.4). A30 does **not** independently re-derive Φ_{face} or the II₁ type.

7.2 Theorem A30.9 — the top-form lives on the complement (selection upgraded)

Why is the vacuum top-form supported on P_{Λ} (rank 83) and not on P_m (rank 38)? The matter carriers P_b (rank 6) and P_c (rank 32) carry the timelike Brown–Kuchař particle current (theorem-forced, ZS-A19/A20). Under an **Exhaustive-Matter-Content axiom** — that the matter content is exactly $P_b + P_c$, with no further particle current — the complement P_{Λ} is the kernel of the matter current and supports no timelike direction. The covariant field with no local propagating degrees of freedom and a forced on-shell constancy is the **top-form** (4-form $F_4 = dA_3$); placing it on the current-free complement gives, by the Henneaux–Teitelboim theorems (IMPORTED-PROVEN), $\partial_{\mu\rho} P_{\Lambda} = 0$, $T_{\mu\nu} = -\rho_{\Lambda} g_{\mu\nu}$, $w = -1$. This upgrades the Top-Form-Selection from HYPOTHESIS-strong to

DERIVED-CONDITIONAL on the matter-content axiom — modest, since nothing strictly forbids a top-form on P_m , but the current-free complement is its natural home. [7.2: DERIVED-CONDITIONAL on Exhaustive-Matter-Content; $w = -1$ IMPORTED-PROVEN.]

7.3 The terminal residual: Unified-Normalization is the cosmological coincidence problem

With Φ_{face} constructed (7.1), the equal weight derived (§5), the unital dynamics proved (§3), and the top-form selected (7.2), one node remains — and it is terminal. The **occupation** fraction is the time-independent rank/trace fact $\tau(q_{\Lambda}) = 83/121$. The **energy** fraction is the cosmological observable

$$\Omega_{\Lambda}(a) = \rho_{\Lambda} / (\rho_{\Lambda} + \rho_{\{m,0\}} a^{-3}), \quad \rho_{\Lambda} = \text{const},$$

which is **time-dependent**: it equals 83/121 at exactly one epoch (today, $a = 1$) and tends to 1 as $a \rightarrow \infty$. So the identity “occupation = energy,” i.e. **Unified-Normalization**, can hold on at most one hypersurface — it is **not** a local identity (A28 retracts the local form). Nor does the de Sitter holographic-equipartition route of v1.1–v1.2 evade it: the maximally-mixed state with 83 vacuum and 38 matter slots describes matter–vacuum **coexistence**, which is the present epoch, not the empty far-future de Sitter horizon (where $\Omega_{\Lambda} \rightarrow 1$ and the 38 matter slots carry vanishing energy). Both the top-form route and the holographic route therefore deliver a **present-epoch** value. UN is precisely the **cosmological coincidence problem** — why ρ_{Λ} and ρ_m are comparable now — which ZS-A28/A29 independently register as a No-Go (the present vacuum normalization is a free integration constant; equivalently a free rank-83 flux vector). [7.3: UN is genuinely OPEN — the coincidence problem; the energy fraction is present-epoch, not unconditional.]

§8. The doubled-semigroup lift, the BRST trace gate, and the classified coincidence escapes

This section adds four results that strengthen the scaffolding around the (unchanged) flagship. None moves the energy fraction off the wall of §7.3; their value is to complete the occupation dynamics, attack the two hardest gates (PT and TFS) with classification rather than analogy, and turn the coincidence wall into a proved No-Go with a finite, classified set of escapes.

8.1 Theorem A30.10 — the exact doubled-semigroup lift (121-dimensional occupation dynamics)

Theorem A30.5 proved the 11-microstate A23 generator unital and primitive with unique stationary $I_{11}/11$ and gap $2A/Q$. The face carrier is the doubled register $V_{11} \otimes V_{11}$ (dimension 121). Lift the generator additively,

$$L_{121} = L_{11} \otimes \text{id} + \text{id} \otimes L_{11} \quad \text{on} \quad M(V_{11} \otimes V_{11}), \quad \text{with} \quad e^{\{tL_{121}\}} = e^{\{tL_{11}\}} \otimes e^{\{tL_{11}\}}.$$

By standard tensor-product quantum-dynamical-semigroup theory the product of two CPTP semigroups is CPTP; a tensor sum of two **unital primitive** generators is unital and primitive, so L_{121} has a unique faithful stationary state, and its spectrum is the set of pairwise sums $\{\lambda_i + \lambda_j\}$ of the L_{11} spectrum. Hence the unique stationary state is $(I_{11}/11) \otimes (I_{11}/11) = \mathbf{I}_{121/121}$ (the maximally-mixed state on the face Hilbert space), and the gap is the smallest nonzero $|\lambda_i + \lambda_j| = 0 + 2A/Q = 2A/Q$. For any rank- r face projector, $\text{Tr}(P_r I_{121}/121) = r/121$, so $(\pi_b, \pi_c, \pi_{\Lambda}) = (6, 32, 83)/121$ are the **exact doubled attractor** of the 11-state parent, not the output of a hand-built 2-state generator. (Verified: the kron-sum spectral identity, the unique zero mode, and the gap $2A/Q$.) This is a **second, finite-dimensional route** to the maximally-mixed occupation that ZS-A24 reaches tracially; it does **not** close ZS-A24’s OPEN continuous-core dynamical lift (F-A24.9), which is infinite-dimensional. [8.1: DERIVED; occupation now has an explicit 121-dimensional primitive unital dynamics.]

8.2 Theorem A30.11 — PT closed to Case A by a witness functional (the corrected argument)

PT is whether the denominator is 121 or 120. The corpus pins it (ZS-A28 §12): the **+1 in 83 = 82 + 1 is the Z_2 -odd mode** removed from CDM ($33 \rightarrow 32$); Case A (physical background) $\rightarrow 121$, $q_{\Lambda} = 83/121$;

Case B (pure-gauge) $\rightarrow 120$, $q_\Lambda = 82/120$. The v1.5 argument — ‘the ZS-A7 seam witness is Hadamard-measurable, hence physical’ — was a **non-sequitur**: measurability of an operator expectation does not by itself place it in BRST cohomology. v1.6 gives the correct argument, a **witness functional**. The seam carries a Z_2 grading β (the trace-preserving order-2 automorphism of ZS-F23, $\beta^2 = 1$, $\tau \circ \beta = \tau$, $\beta \sigma_t = \sigma_t \beta$ — here β is the 2π -rotation operator carrying the $j = 1/2$ sign). Define the **β -twisted (signed) trace**

$$W := \text{STr}_\beta(\cdot) = \tau(\beta \cdot), \quad \text{with BRST } s \text{ acting inner-ly, } s(X) = [Q, X] \text{ (graded commutator, } Q \text{ the odd ghost-number-1 charge).}$$

Two facts close the gate. (i) $W \circ s = 0$: a supertrace annihilates graded commutators, $\text{STr}_\beta([Q, X]) = 0$ for every parity-definite X (the standard identity $\text{STr}(ab) = (-1)^{|a||b|} \text{STr}(ba)$; verified on a finite Z_2 -graded model for both even and odd X). So W is a BRST-invariant (closed) functional. (ii) $W(\hat{u}_{\text{seam}}) \neq 0$: the ZS-A7 signed seam witness $\hat{u}_{\text{seam}}$ is precisely the 2π -odd observable ($\hat{u}_{\text{seam}}(\theta+2\pi) = -\hat{u}_{\text{seam}}(\theta)$), so its β -twisted trace — which projects onto the odd sector — is its defining nonzero invariant. Now suppose the mode were gauge-exact, $\hat{u}_{\text{seam}} = s\chi = [Q, \chi]$. Then $W(\hat{u}_{\text{seam}}) = \text{STr}_\beta([Q, \chi]) = 0$, contradicting (ii). Therefore

$$\hat{u}_{\text{seam}} \text{ is BRST non-exact} \Rightarrow \text{the } Z_2\text{-odd seam mode is a physical background mode (Case A)} \Rightarrow \text{PT} = 121, \quad q_\Lambda = 83/121.$$

This is the argument the review demanded: not ‘measurable,’ but ‘paired nontrivially with a closed witness functional,’ which is exactly what non-exactness requires. [8.2: the witness lemma and $W \circ s = 0$ are PROVEN; $\text{PT} = 121$ (Case A) is **DERIVED-CONDITIONAL** on three named, corpus-grounded identifications — BRST is inner ($s = [Q, \cdot]$), β is the ZS-F23 seam grading, and $\hat{u}_{\text{seam}}$ is the ZS-A7 representative of the disputed mode. The residual is no longer a logical gap but these identifications; an explicit Z_2 -graded ghost complex (the A20/A21 seam cohomology) would discharge them. Not asserted as unconditionally CLOSED.]

8.3 Theorem A30.12 — nonpropagating vacuum-carrier uniqueness (the top-form, by classification)

Theorem A30.9 selected the top-form on the current-null complement as “natural.” This upgrades that to a classification. A massless Abelian p -form gauge potential in four dimensions has

$$N_{\text{dof}}(p) = C(4-2, p) = C(2, p): \quad p = 0 \rightarrow 1, \quad p = 1 \rightarrow 2, \quad p = 2 \rightarrow 1, \quad p = 3 \rightarrow 0, \quad p \geq 4 \rightarrow 0.$$

Demanding a carrier that is (i) local and Lorentz/diffeomorphism covariant, (ii) an Abelian p -form gauge potential, (iii) with **nonzero** gauge-invariant field strength, (iv) with **no local propagating degree of freedom**, and (v) with no extra scalar potential or mass parameter, the only solution is **$p = 3$** : a 3-form potential A_3 with top-form field strength $F_4 = dA_3$ (the 2-form has one dof; the 4-form has no nonzero strength in 4D). Then $F_{\{\mu\nu\rho\sigma\}} = f_{\{\mu\nu\rho\sigma\}}$ gives $T_{\{\mu\nu\}} = -\rho_\Lambda g_{\{\mu\nu\}}$, $w = -1$ (Henneaux–Teitelboim, IMPORTED-PROVEN). So, **given EMC** (matter = $P_b + P_c$, hence $P_\Lambda = \ker J_{\text{matter}}$) **and the carrier axioms (i)–(v)**, $\text{supp } F_4 = P_\Lambda$ is forced — the Top-Form-Selection is no longer “natural” but the unique admissible carrier. [8.3: dof classification PROVEN; $\text{supp } F_4 = P_\Lambda$ DERIVED-CONDITIONAL on EMC + axioms (i)–(v).]

8.4 Theorem A30.13 — the coincidence wall as a No-Go, and its two classified escapes

No-Go (PROVEN). If dust and vacuum are separately conserved ($\dot{\rho}_m + 3H\rho_m = 0$, $\dot{\rho}_\Lambda = 0$) with $w_\Lambda = -1$, then $\Omega_\Lambda(a) = \rho_\Lambda/(\rho_\Lambda + \rho_{\{m,0\}}a^{-3})$ increases monotonically and equals 83/121 on **at most one** hypersurface (verified: exactly one crossing). The flagship energy fraction is therefore a present-epoch boundary value under strict conservation. **Within this declared two-component GR framework** — fixed ranks, dust matter, a top-form vacuum — the exits divide into **two classes**, and we classify both. (This is a statement about the declared framework, not a claim that no other escapes exist: modified gravity, evolving ranks, $w(a) \neq -1$, or additional fluids would each open further routes.)

Escape 1 — interacting rank attractor (a falsifiable branch, NOT a closure). Allow vacuum–matter energy exchange, $\dot{\rho}_m + 3H\rho_m = Q$, $\dot{\rho}_\Lambda = -Q$. Writing the rank-weighted master equation on e-fold time $N = \ln a$ gives $x' = (A/Q)[83(1-x) - 38x] = QA(83/121 - x)$ with $x = \Omega_\Lambda$ and $QA = Q \cdot A = 385/437 \approx 0.881$, a linear relaxation to 83/121; a minimal nonlinear completion $x' = QA \cdot x(1-x)(83/121 - x)$ has fixed points $\{0, 83/121, 1\}$ with **83/121 the unique stable attractor** ($f'(83/121) < 0$; verified). So 83/121 can be a late-time attractor — **but** this requires $\dot{\rho}_\Lambda \neq 0$, i.e. the vacuum is **not** a pure cosmological constant, directly contradicting the $w = -1$ top-form of §8.3 (the ZS-A28 §16.3 structural dilemma: a separate conserved carrier gives $w = -1$ but frees the ratio; an interacting carrier fixes the ratio but is not a top-form). The interaction Q is moreover reverse-engineered from the desired attractor, and the model gives up strict de Sitter ($\Omega_\Lambda \rightarrow 83/121$, not 1). It is a sharp, falsifiable alternative cosmology (testable against CMB/BAO/SN and perturbative stability), **not** a derivation of the coincidence. [Escape-1: HYPOTHESIS — a falsifiable branch, mutually exclusive with the top-form.]

Escape 2 — rank-weighted maximum-entropy slice (a selection principle, NOT a dynamics). Keep $w = -1$, $\dot{\rho}_\Lambda = 0$, and instead define the present. The coarse-grained entropy of distributing energy fraction x over 83 vacuum slots and $(1-x)$ over 38 matter slots, uniform within each block, is $S_{cg}(x) = -x \ln(x/83) - (1-x) \ln((1-x)/38)$. Then $dS_{cg}/dx = \ln[83(1-x)/(38x)]$ vanishes at $83(1-x) = 38x$, i.e. at **83/121**, and $S_{cg}'' = -1/x - 1/(1-x) < 0$, so it is the unique global maximum (verified). Thus 83/121 is the **unique maximum-rank-entropy coexistence slice** — equivalently, the epoch at which the time-dependent energy fraction crosses the time-independent maximally-mixed occupation. Defining $\Sigma_{now} := \operatorname{argmax}_\Sigma S_{cg}(\Sigma)$ makes UN a global **selection principle**, not a local identity. **But** it does not say why an observer is conditioned on the maximum-entropy slice; that needs an observer/typicality measure (e.g. a temporal POVM, the Brown–Kuchař relational clock of ZS-F30, or A24’s modular H-theorem). It sharpens “why now” to “why at maximum rank-entropy,” and does not solve it. [Escape-2: HYPOTHESIS-strong — a variational characterization; observer-conditioning OPEN.]

Net of §8.4. The coincidence wall is a **proved No-Go**, and within the declared two-component framework its exits fall into **two classes**: change the conservation law (Escape 1, at the cost of the top-form and de Sitter) or add a selection measure (Escape 2, at the cost of an observer/exchangeability axiom). This two-class split is **not new** — ZS-A28 §16.7 already proves the ‘top-form XOR single-trace 38:83’ dilemma (a single conserved carrier cannot both give $w = -1$ and fix the energy ratio); Branch A/B is the **dynamical form** of that PROVEN static dilemma, credited here, not re-claimed. Neither escape is a parameter-free derivation; broader frameworks admit further escapes. [8.4: No-Go PROVEN; the two-class split is within the declared framework; the coincidence remains OPEN; the dichotomy is ZS-A28’s.]

8.5 Theorem A30.14 — entropic-typicality selection, derived from exact rank-multinomial counting

Escape 2 said 83/121 maximizes the rank-weighted coarse entropy. v1.6 derives the selection measure rather than positing it. Posit one combinatorial axiom — **rank-multinomial exchangeability**: the universe’s energy budget is N_{eff} exchangeable quanta, each occupying one of the 121 equal-weight rank slots (83 vacuum, 38 matter), so the vacuum count is

$$K_{\Lambda} \sim \text{Binomial}(N_{\text{eff}}, 83/121), \quad E[K_{\Lambda}/N_{\text{eff}}] = \mathbf{83/121} \text{ (exact)}, \quad \text{Var}(K_{\Lambda}/N_{\text{eff}}) = (83 \cdot 38)/(121^2 N_{\text{eff}}).$$

By Stirling, the log-likelihood is $N_{\text{eff}} S_{\text{cg}}(x) + O(\ln N_{\text{eff}})$ with $S_{\text{cg}}(x) = -x \ln(x/83) - (1-x) \ln((1-x)/38)$, so the previously-posed measure $P(x) \propto \exp(N_{\text{eff}} S_{\text{cg}}(x))$ is **not an ansatz but the large-N limit of exact multinomial counting**. The two pieces have sharply different status. (i) The **expected/modal** vacuum fraction is **exactly 83/121, parameter-free** — it is the multinomial mean, equivalently the equipartition (maximally-mixed) point, the same trace fact as the occupation. (ii) The **spread** $\sigma^2 = (83 \cdot 38)/(121^2 N_{\text{eff}}) = 1/(N_{\text{eff}} |S_{\text{cg}}''|)$ carries the one free input N_{eff} — the number of energy quanta — and matches the entropy curvature $S_{\text{cg}}'' = -121^2/(83 \cdot 38) = -4.642$ in the large-N limit (verified to numerical equality, C20/C23). For $N_{\text{eff}} \in [1, 121]$, $\sigma \in [0.46, 0.042]$, and the observed $\Omega_{\Lambda} \approx 0.6847$ sits within $\leq 0.03\sigma$ of 83/121. So the modal observed Ω_{Λ} is 83/121 as a **counting** statement; the **physical** content of the axiom — that energy quanta are exchangeable over the rank slots — is exactly the **rank-to-energy map (= UN)** in new clothes, which is why this **relocates** the coincidence rather than solving it: ‘why 83/121 now’ becomes ‘why are energy sampling units exchangeable over the 121 rank slots,’ a named, falsifiable axiom that connects to the temporal POVM of ZS-F30, the Brown–Kuchař relational clock, and the modular H-theorem of ZS-A24. [8.5: the peak/mean 83/121 and the variance are PROVEN (exact multinomial); the selection of ‘now’ as the modal epoch is DERIVED-CONDITIONAL on the exchangeability (=UN) axiom; the width is N_{eff} -dependent.]

8.6 A documented non-result — the $\text{Vec}_{\{Z_{11}\}}$ fusion route is excluded

A tempting categorical route would read the 11 boundary carriers of B_Z as the 11 elements of the cyclic group Z_{11} (the only order-11 group, since 11 is prime), making $B_Z \simeq \text{Vec}_{\{Z_{11}\}}^{\omega}$ a **pointed** category whose Drinfeld center has exactly $11 \times 11 = 121$ simple objects of quantum dimension 1 — a clean categorical origin for $121 = Q^2$. **This route is false, and recording why sharpens the boundary.** A pointed category’s simple objects are all **invertible** (Frobenius–Perron dimension 1). The Z-sector carriers are the **three** sector irreps of dimensions (3, 2, 6) (ZS-U5: $Q = 11$ counts sector SLOTS as irrep dimensions, **not** group elements), so B_Z has three simple objects of FPdim 3, 2, 6 — not eleven of FPdim 1. Its total FPdim² is $3^2 + 2^2 + 6^2 = \mathbf{49 = 7^2}$ (the operator-sampling normalizer Σd^2 of ZS-A23/A24, the source of the $\omega = (9, 4, 36)/49$ weight), **not** 11. Hence B_Z is **not pointed** and $B_Z \neq \text{Vec}_{\{Z_{11}\}}$. What survives is the weaker, correct statement of §7.1: the **dimension count** $121 = Q^2 = (\dim V_{11})^2$ matches the dimension of ZS-A24’s Turaev–Viro double, with no cyclic-group fusion implied. [8.6: NON-CLAIM / RETRACTED-as-proposed — the $\text{Vec}_{\{Z_{11}\}}$ fusion is excluded by the (3,2,6) FPdim spectrum; only the dimension-count match is retained.]

§9. Falsification gates

Gate	Kind	Trigger
F-A30.5	GENUINE	If the A23-mediator Lindbladian were not unital ($D(I) \neq 0$), or not primitive (more than one zero eigenvalue), $G\text{-}d1$ is not closed.

Gate	Kind	Trigger
F-A30.6	GENUINE	If the occupation-to-energy mechanism failed given equal-face-weight (uniform $T +$ equal entropy per slot not giving equal energy per slot), §4.2 fails.
F-A30.7	GENUINE	If the FHK handle operator were not $\text{diag}(1/\eta_a)$, or the explicit 1-3 Pachner move were invariant for non-uniform weights, Theorem A30.7 fails.
F-A30.8	STRUCTURAL	If the boundary holography were shown to be metric (area-weighted) and triangulation-dependent — i.e. NOT the topological cobordism functor B_Z — the equal-weight conclusion would not apply to Z -Spin.
F-A30.11	GENUINE	If 121 mutually orthogonal trace-1/121 projections could NOT be realized in a Π_1 factor (or if Φ_{face} required M_{11}), Theorem A30.8 fails.
F-A30.12	STRUCTURAL	If the energy fraction $\Omega_{\Lambda}(a)$ were time-INDEPENDENT (UN a local identity), the coincidence framing of §7.3 would be wrong — but $\rho_m \propto a^{-3}$, $\rho_{\Lambda} = \text{const}$ forbid this.
F-A30.9	OBSERVATIONAL	If precision cosmology drove $\Omega_{\Lambda,0}$ incompatibly far from $83/121 = 0.686$, the conditional closure is empirically rejected.
F-A30.13	GENUINE	If the doubled generator $L_{11} \otimes \text{id} + \text{id} \otimes L_{11}$ did NOT have unique stationary $I_{121/121}$ or gap $2A/Q$ (e.g. the kron-sum spectral identity failed), Theorem A30.10 fails.
F-A30.14	STRUCTURAL	If no normalized BRST-invariant state existed, or the disputed $+1$ mode were shown to be a genuine J_Z -odd gauge-exact mode (not the unit class), PT would be 120, not 121.
F-A30.15	GENUINE	If a local, covariant, nonpropagating Abelian gauge carrier OTHER than the 3-form existed in 4D ($N_{\text{dof}}(p,4) \neq C(2,p)$), Theorem A30.12 fails.
F-A30.16	STRUCTURAL	If the interacting model (Escape 1) were presented as a parameter-free derivation rather than a falsifiable branch mutually exclusive with the $w=-1$ top-form, or if S_{cg} (Escape 2) did not peak uniquely at $83/121$, §8.4 is falsified.
F-A30.18	GENUINE	If the L_{11} Lindbladian multiplicities were NOT $(1,80,36,3,1)$ (disagreeing with ZS-A24.I-4), or the doubled spectrum did not convolve to a unique zero with gap $2A/Q$, Theorem A30.10/full fails.
F-A30.19	STRUCTURAL	If the Z_2 -odd seam mode were shown gauge-exact (the ZS-A7 signed seam witness not in $H^0(s)$, or not Hadamard-measurable), PT would be Case B (denominator 120, $82/120$), not Case A.
F-A30.20	GENUINE	If S_{cg} did not peak uniquely at $83/121$, or its curvature were not $-121^2/(83 \cdot 38)$, Theorem A30.14 fails; if A30.14's PARAMETER-FREE peak is read as a solved coincidence (rather than a measure-conditional modal value with an N_{eff} -dependent spread), §8.5 is overclaimed.
F-A30.21	STRUCTURAL	If B_Z were shown pointed (all carriers $\text{FPdim } 1$, $\Sigma d^2 = 11$), the $\text{Vec}_{\{Z_{11}\}}$ route would be revived; the $(3,2,6)$ FPdim spectrum ($\Sigma d^2 = 49$) falsifies it as proposed.
F-A30.17	ANTI-OVERCLAIM	If v1.6 is read as moving the flagship $83/121$ ENERGY fraction off A29's wall, as independently re-deriving ZS-A24's Φ_{face}/Π_1 or ZS-A28's dilemma, as solving the coincidence problem (the witness functional closes PT, not the coincidence; the multinomial relocates UN, not removes it), or as establishing a $\text{Vec}_{\{Z_{11}\}}$ fusion, it is falsified by §7.1, §8.2, §8.4–§8.6, and this paper's scope. v1.6 completes the INTERNAL program; the flagship requires EXTERNAL review.

§G. Open Gates Registry (updated v1.6)

Gate	Status in v1.6	Promotion path
G-d1 unital dynamics	CLOSED (Theorem A30.5): the A23-mediator Lindbladian is unital + primitive (unique $I_{11}/11$, gap $2A/Q$).	— (discharged)

Gate	Status in v1.6	Promotion path
G-d121 doubled lift	DERIVED (Theorem A30.10): the additive lift $L_{11} \otimes \text{id} + \text{id} \otimes L_{11}$ is unital + primitive on the 121-dim face Hilbert space, unique stationary $I_{121/121}$, gap $2A/Q$ (kron-sum spectrum).	Does NOT close ZS-A24's continuous-core dynamical lift (F-A24.9), which is infinite-dimensional.
G-g2 equal-face-weight	DERIVED (Theorem A30.7): explicit Pachner 1-3 state-sum invariance forces uniform weights; metric alternative excluded. Executes ZS-A24's deferred state-sum.	— (discharged; conditional on B_Z below)
G- Φ face embedding	CREDITED to ZS-A24 (§7.1): Π_1 trace-1/121 embedding, $\tau(q_a)=1/121$, coarse $(6,32,83)/121$, via $\text{III}_1 \rightarrow \Pi_\infty \rightarrow \Pi_1$. A30 records $\dim 121=Q^2=TV$ double and the executed equal weight; it does NOT independently re-derive Φ_{face} .	Whether Φ_{face} is the CANONICAL B_Z double (uniqueness) — NOT a $\text{Vec}_{\{Z_{11}\}}$ fusion (NON-CLAIM: 11 = irrep dims, not a group order).
G-C Condition C	The Π_1 factor = gravitational-subregion identification (de Sitter-thermal), corpus-wide.	External crossed-product / CLPW construction (ZS-F23 O-F19.6).
G-TFS top-form selection	CLASSIFIED (Theorem A30.12): among local covariant nonpropagating Abelian carriers with nonzero strength, the 3-form $(F_4, w=-1)$ is UNIQUE (p-form dof $C(2,p)$); DERIVED-CONDITIONAL on EMC + carrier axioms (was “natural” in v1.3).	Forbid a top-form on P_m intrinsically (remove the EMC-axiom dependence).
G-PT physical trace	CLOSED to Case A via a witness functional (Theorem A30.11, v1.6): the β -twisted trace $W = \text{STr}_\beta$ satisfies $W \circ s = 0$ (supertrace kills graded BRST commutators) and $W(\hat{u}_{\text{seam}}) \neq 0$, so $\hat{u}_{\text{seam}}$ is BRST non-exact $\rightarrow$ the Z_2 -odd seam mode is physical $\rightarrow$ denominator 121 ($q_\Lambda = 83/121$). This replaces v1.5's ‘measurable $\Rightarrow$ physical’ non-sequitur. DERIVED-CONDITIONAL on BRST-innerness + β = seam grading + $\hat{u}_{\text{seam}}$ = ZS-A7 witness.	Explicit Z_2 -graded ghost complex (A20–A21 seam cohomology) discharging the three identifications.
G-UN coincidence	TERMINAL OPEN = the coincidence problem; proved as a No-Go with TWO classified escapes (A30.13); the Branch-A/B dichotomy is ZS-A28 §16.7's PROVEN dilemma in dynamical form (credited). A30.14 upgrades Escape-2: 83/121 is the MODAL Ω_Λ under entropic typicality (peak parameter-free, spread N_{eff} -dependent).	EXTERNAL adversarial review (A28 §17): operator-algebra check of Π_1/TV -double; cosmologist framing; or a derivation of the typicality measure / dark-sector interaction.
G-cat $\text{Vec}_{\{Z_{11}\}}$	RETRACTED-as-proposed (§8.6): B_Z carries $\text{FPdim}(3,2,6)$, $\Sigma d^2=49=7^2 \neq 11$, so B_Z is NOT pointed and $\neq \text{Vec}_{\{Z_{11}\}}$. Only the dimension-count $121=Q^2=\dim(TV \text{ double})$ survives.	— (closed-negative as proposed; the canonical-double identity remains the Φ row above).

§10. Cross-paper dependency

Table 3. Dependency audit. v1.4 imports and elevates; it does not modify upstream locked results, and it credits ZS-A24 for Φ_{face}/Π_1 .

Result	Source / status	Use in v1.4
dimension-weighted mediator law $q=\kappa^2 d_j$; $\pi_i=d_i/D$	ZS-A23.14a, A23.5a (PROVEN)	edge-isotropy $\rightarrow$ unital (§3); doubled lift (§8.1)
unital $\rightarrow I/N$; primitive $\rightarrow$ unique steady state	Nielsen-Chuang; Evans 1977 (IMPORTED)	G-d1 closure (§3)

Result	Source / status	Use in v1.4
tensor-product QDS: $e^{\{t(A \otimes I + I \otimes A)\}} = e^{\{tA\}} \otimes e^{\{tA\}}$; spec = pairwise sums	standard semigroup theory (IMPORTED-PROVEN)	doubled-semigroup lift A30.10 (§8.1)
$\Phi_{\text{face}} : \mathbb{C}^{121} \rightarrow \text{II}_1 \text{ corner}$ ($\tau=1/121$, coarse (6,32,83)/121); $\text{III}_1 \rightarrow \text{II}_\infty \rightarrow \text{II}_1$	ZS-A24 §10–§13 v1.5/v2.x (DERIVED-COND; finite route RETRACTED)	credited, not re-derived (§7.1); II_1 home (NOT cardinality)
de Sitter = Type II_1 , max-entropy tracial; combinatorial holography via BV–BFV	CLPW; ZS-F23/A24/F0 §8.5 (IMPORTED/DERIVED)	attractor (§4.1); equal weight (§5)
2D state-sum TQFT $\leftrightarrow$ special Frobenius; Pachner $\Rightarrow$ separable	Fukuma–Hosono–Kawai; Turaev–Viro (IMPORTED-PROVEN)	executes A24-deferred state-sum (§5)
top-form / 4-form $F_4 \Rightarrow w=-1$; p-form dof = $C(D-2,p)$	Henneaux–Teitelboim; Hodge dof count (IMPORTED-PROVEN)	top-form carrier uniqueness A30.12 (§8.3)
BRST cohomology: physical = s-closed mod s-exact	Henneaux–Teitelboim (quantization); Kugo–Ojima (IMPORTED-PROVEN)	unit-survival lemma A30.11 (§8.2)
coincidence No-Go (free vacuum normalization / flux); interacting dark energy	ZS-A28/A29; Wetterich/Amendola-type scaling (PROVEN/IMPORTED)	No-Go + escapes A30.13 (§8.4)
(6,32,83)/121 budget	ZS-F2/A19/A20/Lemma 16.1 (PROVEN)	inputs (§2)

§11. Conclusion

v1.4 strengthens the scaffolding around the (unchanged) flagship and corrects the ZS-A24 boundary. Four results are added: the doubled-semigroup lift $\mathbf{L}_{121} = \mathbf{L}_{11} \otimes \text{id} + \text{id} \otimes \mathbf{L}_{11}$ gives the 121-dimensional occupation an explicit unital primitive dynamics with unique stationary $\mathbf{I}_{121/121}$ and gap $2A/Q$ (Theorem A30.10 — a finite-dimensional SECOND route to ZS-A24’s maximally-mixed corner, NOT a closure of A24’s infinite-dimensional continuous-core lift F-A24.9); a BRST unit-survival lemma shows the unit/background class survives BRST whenever a normalized physical state exists, so $\text{PT} = 121$ if the disputed +1 mode is that class (Theorem A30.11); a p-form degree-of-freedom classification shows the 3-form (F_4 , $w = -1$) is the UNIQUE local covariant nonpropagating Abelian vacuum carrier, upgrading the Top-Form-Selection from ‘natural’ to a classification theorem (Theorem A30.12); and the coincidence wall is proved as a No-Go with two fully classified escapes (Theorem A30.13).

Two corrections discharge the external critique. The face embedding Φ_{face} and its Type II_1 home are **ZS-A24’s**, reached there via the genuine Type $\text{III}_1 \rightarrow \text{II}_\infty \rightarrow \text{II}_1$ corner (Takesaki / Witten / CLPW); v1.3’s cardinality phrasing ($121 > 11 \Rightarrow \text{II}_1$) was a non-sequitur and is replaced by A24’s construction, which A30 now **credits rather than re-derives**. A30’s genuine increment over A24 is therefore explicit and narrow: it executes A24’s **deferred** Pachner/FHK state-sum (with the metric alternative positively excluded), supplies the finite-dimensional doubled-semigroup dynamics, and assembles the occupation→energy budget. The speculative $\text{Vec}_{\{Z_{11}\}}$ fusion route is recorded as a **NON-CLAIM**, since the 11 register slots are irrep dimensions ($3 + 2 + 6$), not the elements of an order-11 group.

The honest net result, after seven versions, is unchanged where it matters: **$\Omega_{\mathbf{A},0} = 83/121$ is an unconditional OCCUPATION and a present-epoch ENERGY fraction.** v1.6 does not move that flagship; it closes the two residuals the external review flagged and repairs the contradictory checks. The

physical-trace gate PT is closed to **Case A** by the **correct** argument — not the v1.5 ‘measurable, therefore physical’ non-sequitur, but a **witness functional**: the β -twisted (signed) trace $W = \text{STr}_\beta$ satisfies $W \circ s = 0$ (a supertrace annihilates graded BRST commutators) and $W(\hat{u}_{\text{seam}}) \neq 0$, so $\hat{u}_{\text{seam}}$ is BRST non-exact and the Z_2 -odd seam mode is physical (denominator 121). The entropic-typicality selection is derived from **exact rank-multinomial counting**: $K_\Lambda \sim \text{Binomial}(N_{\text{eff}}, 83/121)$ has mean 83/121 and a Stirling limit reproducing the entropy curvature, so the $\exp(N_{\text{eff}} S_{\text{cg}})$ measure is counted rather than posited — the peak 83/121 is parameter-free, the width scales as $N_{\text{eff}}^{-1/2}$, and the exchangeability axiom is exactly the rank-to-energy map UN, so the construction relocates the coincidence rather than solving it. The doubled-semigroup multiplicities (1,80,36,3,1) now carry an **analytic proof** (coherence eigenvalues $-\frac{1}{2}\gamma(d_i+d_j)$ on the degree-(2,9) graph plus the population spectrum), upgrading C19 from a numeric match to a theorem.

Four repairs accompany the two closures: the broken finite-complex check that returned $\dim H^0 = 0$ while asserting survival is replaced by the witness gate (C14); the $\text{Vec}_{\{Z_{11}\}} / \text{Dijkgraaf-Witten } Z_{11} \times Z_{11}$ ‘translation-transitivity’ gloss is removed everywhere (it conflicted with B_Z carrying $\text{FPdim}(3,2,6)$, $\Sigma d^2 = 49 \neq 11$; the equal weight rests on Pachner invariance alone); the face-algebra notation is regularized to $D_{121} \simeq \mathbb{C}^{121} \subseteq M_{121}$ (Lindbladian on the full matrix algebra, face idempotents on the diagonal); and ‘Zero Free Parameters’ is restated as ‘zero new fitted Z-Spin parameters, structural conditions explicit, one open present-epoch normalization.’ With PT closed to Case A, the typicality measure derived, the spectrum proved, the No-Go and its two-class escape set classified, and the review defects fixed, the **internal program is complete**. The energy fraction remains DERIVED-CONDITIONAL on $\{\text{UN}, \text{TFS}, B_Z\text{-double}, \text{PT}\}$; the terminal node UN is the cosmological coincidence problem; and — per the corpus’s own repeated discipline (A28 §17, A29) that internal iteration converges to honesty but not to closure — the genuine next step is **external adversarial review**: an operator-algebraic audit of the $\Pi_1 / \text{Turaev-Viro double}$, the doubled-semigroup lift, and the seam BRST complex, and a cosmologist’s scrutiny of the coincidence framing and the rank-multinomial measure. This is the terminal internal version. Zero new fitted Z-Spin parameters.

Acknowledgements & Code Availability

v1.6 closes review items 5 and 6 with the correct arguments (the PT witness functional and the exact rank-multinomial typicality), adds the analytic spectrum proof, and repairs the contradictory checks (the broken $\dim H^0$ toy, the $\text{Vec}_{\{Z_{11}\}} / \text{DW } Z_{11}$ gloss, the face-algebra notation, the parameter wording). The verification script `zs_a30_verify_v1_4.py` adds six CORE checks (A30.10 doubled lift: unique $I_{121}/121$, gap $2A/Q$ via the kron-sum spectral identity; A30.11 BRST unit-survival + finite-complex $\dim H^0$; A30.12 p-form dof $C(2,p)$ with the 3-form unique; A30.13 No-Go single-crossing, Escape-1 attractor fixed points and stability, Escape-2 entropy maximum at 83/121), one IMPORTED check (tensor-product semigroup), and two SCOPE audits (A24 overlap acknowledged; escapes classified, neither solves, external review next), on top of the v1.0–v1.4 categories. The v1.6 script (`zs_a30_verify_v1_6.py`) strengthens C19 (ANALYTIC decomposition — coherence $-\frac{1}{2}\gamma(d_i+d_j)$ on the degree-(2,9) graph + population $\rightarrow (1,80,36,3,1)$), C20 (entropic typicality via EXACT binomial mean/variance), and adds C22 (the PT witness identity $W \circ s = 0$, $W(\hat{u}_{\text{seam}}) \neq 0$) and C23 (binomial variance = entropy curvature for all N_{eff}); it FIXES the broken C14 (now the witness gate, $\dim H^0 = 1$ with the surviving class detected by the cocycle) and I6 (FHK handle operator $H = I$ only, the Z_{11} -pointed claim removed), and unifies the docstring/version to v1.6. It carries no fail-open clause and exits non-zero on any inconsistency. (A, Q, $\dim Z$) LOCKED.

Appendix A. The unital construction

On the 11 microstates ($X = \{0,1,2\}$, $Z = \{3,4\}$, $Y = \{5,\dots,10\}$), the jump set is $\{L_{\beta\alpha} = \sqrt{\gamma} |\beta\rangle\langle\alpha| : (\alpha,\beta) \text{ adjacent on } X\text{--}Z\text{--}Y, \text{ both orderings}\}$, $\gamma = \kappa^2(X\text{--}Y \text{ absent})$. Unitality: each unordered adjacent pair contributes $\gamma(|\beta\rangle\langle\alpha|, |\alpha\rangle\langle\beta|) + (|\alpha\rangle\langle\beta|, |\beta\rangle\langle\alpha|) =$

$\gamma(|\beta\rangle\langle\beta| - |\alpha\rangle\langle\alpha| + |\alpha\rangle\langle\alpha| - |\beta\rangle\langle\beta|) = 0$, so $D(I) = 0$. The vectorized Lindbladian $\mathcal{L} = \Sigma\gamma(L \otimes \bar{L} - \frac{1}{2} L \dagger L \otimes I - \frac{1}{2} I \otimes (L \dagger L)^T)$ has exactly one zero eigenvalue; its kernel is $\text{vec}(I/11)$; the largest nonzero real part is $-2A/Q$ (the A23 slow mode). Summing the per-microstate rate γ over the d_j targets on an edge gives the A23 sector rate $q_{\{i \rightarrow j\}} = \gamma d_j$: $q_{\{X \rightarrow Z\}} = 2\kappa^2 = 2A/Q$, $q_{\{Z \rightarrow Y\}} = 6\kappa^2 = 6A/Q$. [IMPORTED-PROVEN edge-isotropy (A23); the unital/primitive conclusions are PROVEN here.]

Appendix B. Numerical verification

CORE-THEOREM. (G-d1) 11-microstate A23 Lindbladian: $D(I) = 0$ to machine precision; $\Sigma\gamma[L, L^\dagger] = 0$; exactly one zero eigenvalue; stationary state $= I/11$; coarse occupations $(3,2,6)/11$; spectral gap $-2A/Q = -0.01456$. (occupations) 2-state stationary $(38/121, 83/121)$; 3-block $(6,32,83)/121$; $\text{Tr}(P_i I/121) = \text{rank}_i/121$. (G-g2) $I/121$ is the unique maximum-entropy state; equal energy per slot $\rightarrow \Omega_\Lambda = 83/121$. (A30.7 / TQFT) FHK handle operator $H = \text{diag}(1/\eta_a) = I$ iff η uniform; explicit 1-3 Pachner move on $\mathbb{C}^3$ — $Z(\text{tetrahedron}, 4 \text{ triangles}) = Z(1\text{-}3 \text{ subdivision}, 6 \text{ triangles}) = 3.000$ for uniform η , $3.485 \neq 4.212$ for non-uniform; genus formula $Z(\text{sphere}) = 121$ (uniform) vs 133 (metric) — metric excluded. [The $Z_{11} \times Z_{11}$ translation gloss is RETRACTED in v1.6; equal weight rests on Pachner invariance.] (A30.8 / Φ_{face}) 121 mutually orthogonal idempotents with $\tau = 1/121$ each, summing to 1 ($\Sigma\tau = 1$), realizable in a II_1 factor (continuous trace) but not in M_{11} ($121 > 11$); coarse $\tau(q_b, q_c, q_\Lambda) = (6,32,83)/121$; $\dim 121 = Q^2 = 11^2 = \text{TV-double dimension}$. (A30.8 / UN) $\Omega_\Lambda(a) = \rho_\Lambda/(\rho_\Lambda + \rho_m, 0 a^{-3})$ is time-dependent: $= 83/121$ only at $a = 1$, $\rightarrow 1$ as $a \rightarrow \infty$, while the rank/occupation $83/121$ is time-independent — UN is not a local identity. IMPORTED-PROVEN. A23 sector $(3,2,6)/11$, spectrum $\{0, -2A/Q, -A\}$; unital channel fixes I/N ; Weyl–Heisenberg covariance $\rightarrow I/N$; Gibbons–Hawking $S_{\text{dS}} = 8\pi^2 \bar{M}^2/H^2$; top-form / Henneaux–Teitelboim $\rightarrow w = -1$ (tensor identity $\epsilon_{\{\mu\alpha\beta\gamma\}} \epsilon_{\nu}^{\{\alpha\beta\gamma\}} = -3! g_{\{\mu\nu\}}$ forces $T_{\mu\nu} = -\rho_\Lambda g_{\mu\nu}$). REGRESSION. $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$; ranks $(6, 32, 83)$. NEW v1.4. (A30.10) build the 11-microstate Lindbladian, confirm unique zero mode + gap $-2A/Q$, and the kron-sum identity $\text{spec}(A \otimes I + I \otimes A) = \{a_i + a_j\}$ so the doubled stationary is $I_{121}/121$ with gap $2A/Q$. (A30.11) finite BRST complex $s_0 s_{-1} = 0$, $\dim H^0 = \dim \ker s_0 - \text{rank } s_{-1}$; the lemma (normalized invariant $\Rightarrow [1] \neq 0$) is representation-independent. (A30.12) $N_{\text{dof}}(p, 4) = C(2, p) = (1, 2, 1, 0)$ for $p=0-3$; only $p=3$ has zero dof with a nonzero F_4 . (A30.13) with $Q=0, w=-1$, $\Omega_\Lambda(a)$ crosses $83/121$ exactly once; the nonlinear $x' = QA \cdot x(1-x)/(83/121-x)$ has fixed points $\{0, 83/121, 1\}$ with $f'(83/121) < 0$ (stable), $QA = Q \cdot A = 385/437$; $S_{\text{cg}}(x)$ peaks uniquely at $83/121$ with $S_{\text{cg}}'' < 0$. SCOPE-AUDIT. energy fraction NOT fully unconditional — DERIVED-CONDITIONAL on $\{\text{UN (coincidence problem), TFS (matter-content axiom), } B_Z \text{ double, PT}\}$; $\Phi_{\text{face}}/\text{II}_1$ CREDITED to ZS-A24 (not re-derived; II_1 NOT from cardinality); escapes classified, NEITHER solves the coincidence (Escape-1 contradicts $w=-1$ + reverse-engineers Q ; Escape-2 needs an observer measure); $\text{Vec}_{\{Z_{11}\}}$ fusion is a NON-CLAIM; external adversarial review is the disciplined next step; no DESI claim.

Appendix C. Full derivations (v1.6)

C.1 The doubled-semigroup spectrum — analytic multiplicity proof

The 11-microstate generator can be diagonalised in closed form, upgrading C19 from a tolerance-binned numeric match to a proof. The mediator graph has $X \cup Y = 9$ nodes of degree 2 and $Z = 2$ nodes of degree 9 (each Z couples to all 3 X and all 6 Y ; each X or Y couples to both Z). Write $\gamma = A/Q$ for the per-edge rate. The dissipator splits into a coherence sector and a population sector. For an off-diagonal coherence $|i\rangle\langle j|$ ($i \neq j$) the jump gain vanishes and $\Sigma_k \gamma L_k \dagger L_k = \gamma \cdot \text{diag}(\text{degree})$, so $|i\rangle\langle j|$ is an exact eigenvector with eigenvalue $-\frac{1}{2}\gamma(d_i + d_j)$: $X/Y \leftrightarrow X/Y$ gives $-\frac{1}{2}\gamma(2+2) = -2\gamma = -2A/Q$ with multiplicity $9 \cdot 8 = 72$; $X/Y \leftrightarrow Z$ gives $-\frac{1}{2}\gamma(2+9) = -(11/2)\gamma = -A/2$ (since $Q = 11$) with multiplicity $9 \cdot 2 \cdot 2 = 36$; $Z \leftrightarrow Z$ gives $-\frac{1}{2}\gamma(9+9) = -9\gamma = -9A/Q$ with multiplicity 2. The population sector is the classical generator on 11 nodes, with spectrum $0 (\times 1)$, $-2\gamma (\times 8)$, $-9\gamma (\times 1)$, $-11\gamma = -A (\times 1)$. Summing the two sectors gives exactly $0(1)$, $-2A/Q(72+8 = 80)$, $-A/2(36)$, $-9A/Q(2+1 = 3)$, $-A(1)$ — the multiplicities $(1, 80, 36, 3, 1)$ of ZS-A24.I-4, now with each value tied to its degeneracy by construction (verification C19). [PROVEN.]

C.1b The complete doubled spectrum

The A23-mediator Lindbladian on the 11 microstates, vectorised on M_{11} (a 121-dimensional Liouville space), has spectrum $\sigma(\mathcal{L}_{11}) = \{0, -2A/Q, -A/2, -9A/Q, -A\}$ with multiplicities $(1, 80, 36, 3, 1)$ summing to 121 — reproducing ZS-A24.I-4 by independent construction (verification C19). For the doubled generator $\mathcal{L}_{121} = \mathcal{L}_{11} \otimes \text{id} + \text{id} \otimes \mathcal{L}_{11}$ on $M_{11} \otimes M_{11}$ the semigroup factorises, $e^{\{t\mathcal{L}_{121}\}} = e^{\{t\mathcal{L}_{11}\}} \otimes e^{\{t\mathcal{L}_{11}\}}$, so the spectrum is the multiset of pairwise sums $\{\lambda_i + \lambda_j\}$ with multiplicities given by the self-convolution of $(1, 80, 36, 3, 1)$. The total is $121^2 = 14641$; the zero eigenvalue occurs once (only $0 + 0$, since every other λ has strictly negative real part), giving the unique stationary state $(I_{11}/11) \otimes (I_{11}/11) = I_{121}/121$; and the slowest nonzero mode is $0 +$

$(-2A/Q) = -2A/Q$, so the gap is exactly $2A/Q$. Primitivity is inherited: a tensor sum of primitive generators has a one-dimensional kernel. [PROVEN; C19.]

C.2 The 3-form Hamiltonian constraint count

A massless Abelian p -form A_p in $D = 4$ has $C(4,p) = (1,4,6,4,1)$ components for $p = (0,1,2,3,4)$. Dirac–Bergmann analysis removes, per gauge generation, the temporal components and their conjugate primary constraints together with the secondary (Gauss) constraints and the residual gauge orbit, leaving $C(D-2, p) = C(2,p) = (1,2,1,0)$ physical polarisations. For $p = 3$ all four components of A_3 are non-dynamical (zero propagating d.o.f.), yet the field strength $F_4 = dA_3$ is nonzero; its equation of motion $\partial_\mu(\sqrt{-g} F^{\mu\nu\rho\sigma}) = 0$ forces $F_{\mu\nu\rho\sigma} = f \varepsilon_{\mu\nu\rho\sigma}$ with f constant, whence $T_{\mu\nu} = -\frac{1}{2} f^2 g_{\mu\nu} = -\rho_\Lambda g_{\mu\nu}$, $w = -1$ (Henneaux–Teitelboim). Among all four-dimensional Abelian gauge carriers with nonzero strength and no local mode, $p = 3$ is therefore unique. [PROVEN classification; C21; IMPORTED-PROVEN $w = -1$.]

C.3 Entropic typicality from exact rank-multinomial counting

Model the energy budget as N_{eff} exchangeable quanta, each landing in one of the 121 equal-weight rank slots (83 vacuum). The vacuum count is then $K_\Lambda \sim \text{Binomial}(N_{\text{eff}}, 83/121)$, with EXACT mean $E[K_\Lambda/N_{\text{eff}}] = 83/121$ and variance $\text{Var}(K_\Lambda/N_{\text{eff}}) = p(1-p)/N_{\text{eff}} = (83 \cdot 38)/(121^2 N_{\text{eff}})$. The Stirling expansion of the log-pmf is $\ln P(k) = N_{\text{eff}} \cdot [-x \ln(x/p) - (1-x) \ln((1-x)/(1-p))] + O(\ln N_{\text{eff}})$ with $x = k/N_{\text{eff}}$ and $p = 83/121$; with the rank degeneracies (83 vacuum, 38 matter) folded into the slot weights this is exactly $N_{\text{eff}} \cdot S_{\text{cg}}(x)$ for $S_{\text{cg}}(x) = -x \ln(x/83) - (1-x) \ln((1-x)/38)$. Hence the previously-posed $\exp(N_{\text{eff}} S_{\text{cg}})$ measure IS the large- N limit of exact multinomial counting. The peak is $dS_{\text{cg}}/dx = \ln[83(1-x)/(38x)] = 0 \Rightarrow x^* = 83/121$; the curvature is $S_{\text{cg}}''(x^*) = -(1/x^* + 1/(1-x^*)) = -121^2/(83 \cdot 38) = -4.6420$, and $1/(N_{\text{eff}} |S_{\text{cg}}''|) = (83 \cdot 38)/(121^2 N_{\text{eff}})$ matches the binomial variance identically for every N_{eff} (verified C20/C23). So the peak/mean $83/121$ is parameter-free, the width $\sigma \propto N_{\text{eff}}^{-1/2}$, and the lone physical assumption is exchangeability of energy quanta over the rank slots — i.e. the rank-to-energy map UN , which is why the construction relocates rather than solves the coincidence. [Peak/variance PROVEN (exact multinomial); selection DERIVED-CONDITIONAL on exchangeability ($=UN$); width N_{eff} -dependent. C20/C23.]

C.4 The physical-trace witness functional (PT, Case A)

PT asks whether the disputed Z_2 -odd seam mode is BRST-exact (gauge, denominator 120) or not (physical, denominator 121). The closure is a witness functional, not a measurability claim. Equip the seam algebra with its Z_2 grading β (ZS-F23: $\beta^2 = 1$, $\tau \circ \beta = \tau$, $\beta \sigma_\tau = \sigma_\tau \beta$; the 2π -rotation operator) and let BRST act inner-ly, $s(X) = [Q, X]$ (graded commutator, Q odd). Define $W = \text{STr}_\beta(\cdot) = \tau(\beta \cdot)$. Then (i) $W \circ s = 0$ because a supertrace annihilates graded commutators — $\text{STr}_\beta(QX - (-1)^{|X|} \{X\} Q) = 0$ for every parity-definite X , the identity $\text{STr}(ab) = (-1)^{|a||b|} \text{STr}(ba)$ (verified on a finite Z_2 -graded model for even and odd X , C22); and (ii) $W(\hat{u}_{\text{seam}}) \neq 0$ because the ZS-A7 signed seam witness is the 2π -odd observable that the β -twisted trace is built to detect. If the mode were exact, $\hat{u}_{\text{seam}} = s\chi = [Q, \chi]$, then $W(\hat{u}_{\text{seam}}) = \text{STr}_\beta([Q, \chi]) = 0$, contradicting (ii). Therefore $\hat{u}_{\text{seam}}$ is BRST non-exact $\Rightarrow$ the Z_2 -odd seam mode is physical (Case A) $\Rightarrow PT = 121$. A corrected finite complex $(C^{-1} \rightarrow C^0 \rightarrow C^1$ with $s_0 s_{-1} = 0)$ carries $\dim H^0 = 1$ with the surviving class detected by the cocycle W — replacing the v1.5 toy that returned $\dim H^0 = 0$ while asserting survival (C14 fixed). [Witness lemma and $W \circ s = 0$ PROVEN; $PT = 121$ DERIVED-CONDITIONAL on BRST-innerness, β = seam grading, and $\hat{u}_{\text{seam}}$ = the ZS-A7 representative. C14/C22.]

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Version History

v1.0 (June 2026): Executed the three internally-decidable tasks of ZS-A29 §8.3; elevated the rank-weighted master-equation candidate from HYPOTHESIS-strong to DERIVED-CONDITIONAL with $\Omega_{\Lambda,0} = 83/121$ conditional on $\{G_1, \text{unital/covariant dynamics, de Sitter-thermal, PT}\}$; unified G2 with G3.

v1.1 (June 2026): Closed G-d1 and reduced G-g2. (1) **Theorem A30.5** — the parent Lindbladian built from the ZS-A23 edge-isotropic mediator law is **unital** ($D(I) = 0$ by conjugate-pair cancellation, since edge-isotropy makes microstate rates symmetric) and **primitive** (connected $X \rightarrow Z \rightarrow Y$ path), so $I/121$ is the unique, globally attracting stationary state with gap $2A/Q$; the

v1.0 unital/covariant condition is discharged and the occupations (38/121, 83/121) are DERIVED. (2) **Theorem A30.6** — the attractor is the de Sitter Type II₁ maximum-entropy tracial state (CLPW/F23/A24, DERIVED-CONDITIONAL on Condition C); the occupation-to-energy step closes by horizon thermodynamics (uniform T_{dS}, equal entropy ln 2 per slot at maximum entropy, dE = T dS → equal energy per slot → 83/121) given the single residual, the **face embedding** (equal-face-weight = combinatorial, not metric, holography), which ZS-A24 registers as the deepest gap and which is sharpened here to a decidable TQFT state-sum statement (Pachner invariance, one label per face). Net: $\Omega_{\Lambda,0} = 83/121$ as an OCCUPATION is DERIVED; as an ENERGY fraction it is DERIVED-CONDITIONAL on {face embedding, PT}. Explicitly NOT claimed unconditional. Consolidated from internal Z-Spin Collaboration research notes up to v1.1.0.

v1.2 (June 2026): Executed the TQFT state-sum computation and discharged ZS-A24's deepest gap. **Theorem A30.7** — the equal-face-weight is forced by topological (Pachner) invariance on the abelian face algebra $\mathbb{C}^{121}$. By Fukuma–Hosono–Kawai a triangulation-invariant 2D state sum is a special symmetric Frobenius algebra; the special condition $H = m \circ \Delta = \text{diag}(1/\eta_a) = I$ forces uniform weights. Verified by an explicit brute-force 1-3 Pachner move (tetrahedron 4 triangles vs 1-3 subdivision 6 triangles: $3 = 3$ for uniform, $3.49 \neq 4.21$ for non-uniform) and by the genus formula (sphere amplitude 121 uniform vs 133 metric); [the Dijkgraaf–Witten $Z_{11} \times Z_{11}$ translation gloss is RETRACTED in v1.6 — see §8.6.] Metric (area) weighting is triangulation-dependent and excluded. Net: $\Omega_{\Lambda,0} = 83/121$ is a derived occupation (G-d1) and the equal-face-weight its energy reading needs is now DERIVED; the energy fraction is DERIVED-CONDITIONAL on the reduced set {G1 carrier-selection, the B_Z topological-holography identification, PT}. Explicitly NOT claimed fully unconditional. Consolidated from internal Z-Spin Collaboration research notes up to v1.2.0.

v1.3 (June 2026): Constructed the face embedding explicitly and located the terminal residual. **Theorem A30.8** — $\Phi_{\text{face}} : \mathbb{C}^{121} \rightarrow M_{dS}$ is realized as 121 mutually orthogonal projections of continuous trace 1/121 in the Type II₁ factor (impossible in M₁₁, where $121 > 11$; Murray–von Neumann), with coarse traces (6,32,83)/121 and dimension $121 = Q^2 = 11^2$ matching the Turaev–Viro double of the pointed cobordism functor B_Z; existence and equal weight DERIVED, the canonical-double identification DERIVED-CONDITIONAL. **Theorem A30.9** — given an Exhaustive-Matter-Content axiom, the vacuum top-form is supported on the current-free complement P_Λ and forces $w = -1$ (Henneaux–Teitelboim, IMPORTED-PROVEN); Top-Form-Selection upgraded from HYPOTHESIS-strong to DERIVED-CONDITIONAL. **Terminal residual** — the energy fraction $\Omega_{\Lambda}(a) = \rho_{\Lambda}/(\rho_{\Lambda} + \rho_{m,0} a^{-3})$ is time-dependent and equals 83/121 only at $a = 1$; UN cannot be a local identity, and the de Sitter holographic route gives a present-epoch value too. UN is the cosmological coincidence problem (A28/A29 No-Go). Net: $\Omega_{\Lambda,0} = 83/121$ is an unconditional OCCUPATION and a present-epoch ENERGY fraction, DERIVED-CONDITIONAL on {UN, TFS-axiom, canonical B_Z double, PT}; the irreducible obstruction is located by two independent routes, not removed. Explicitly NOT an unconditional energy fraction. Consolidated up to v1.3.0.

v1.4 (June 2026): Integrated external adversarial review; added genuine math increments and corrected the ZS-A24 boundary. **Theorem A30.10** (doubled-semigroup lift) — $L_{121} = L_{11} \otimes \text{id} + \text{id} \otimes L_{11}$ is unital + primitive on the 121-dimensional face Hilbert space, with unique stationary $I_{121/121}$ and gap $2A/Q$ (the kron-sum spectral identity); a finite-dimensional SECOND route to ZS-A24's maximally-mixed corner, which does NOT close A24's infinite-dimensional continuous-core lift (F-A24.9). **Theorem A30.11** (BRST unit-survival) — a normalized BRST-invariant state forces $[1] \neq 0$ in $H^0(s)$; PT = 121 DERIVED-CONDITIONAL on the +1 mode being the unit/background class. **Theorem A30.12** (top-form carrier uniqueness) — massless p-form dof = C(2,p) in 4D, so the 3-form (F₄, $w = -1$) is the UNIQUE local covariant nonpropagating Abelian vacuum carrier; TFS upgraded from 'natural' to a classification theorem (DERIVED-CONDITIONAL on EMC + carrier axioms). **Theorem A30.13** (No-Go + classified escapes) — with $Q = 0$, $w = -1$, $\Omega_{\Lambda}(a)$ hits 83/121 on at most one slice (PROVEN); Escape-1 (interacting rank attractor $x' = QA \cdot x(1-x)/(83/121-x)$, $QA = 385/437$, 83/121 stable) is a falsifiable branch that contradicts the $w = -1$ top-form and reverse-engineers Q; Escape-2 (rank-weighted entropy max at 83/121) characterizes 'now' but needs an observer measure — NEITHER solves the coincidence. **Corrections (external critique)**: §7.1's II₁ justification is fixed — the cardinality remark ' $121 > 11 \Rightarrow \text{II}_1$ ' was a non-sequitur and is replaced by ZS-A24's genuine $\text{III}_1 \rightarrow \text{II}_{\infty} \rightarrow \text{II}_1$ corner (Takesaki/Witten/CLPW); Φ_{face} and its II₁ home are credited to A24, not re-derived (A30's increment is the executed Pachner/FHK state-sum, the doubled lift, and the occupation→energy assembly); the A24 citation is pinned to v2.0; the speculative $\text{Vec}_{\{Z_{11}\}}$ fusion is recorded as a NON-CLAIM ($11 = 3+2+6$ irrep dims, not a group order); the conclusion's stale v1.2 paragraph is removed and the section numbering (§8–11) is regularized. **Flagship unchanged**: $\Omega_{\Lambda,0} = 83/121$ remains an unconditional OCCUPATION and a present-epoch ENERGY fraction; the No-Go is proved and its escapes exhausted, so the disciplined next step is external operator-algebra / cosmology review (A28 §17), not a v1.5. Verification: CORE-THEOREM 18, IMPORTED-PROVEN 8, REGRESSION 4, SCOPE-AUDIT 8, all consistent. No new fitted parameters. Consolidated up to v1.4.0. (A = 35/437, Q = 11, dim Z = 2) LOCKED.

v1.5 (June 2026): Full derivations + verification of the v1.4 theorems, four sharpenings, and two honesty corrections — completing the internal program. **A30.10/full** — the L_{11} Lindbladian multiplicities (1,80,36,3,1) reproduce ZS-A24.I-4 by independent construction and convolve to the complete 121^2 doubled spectrum (unique zero $\rightarrow I_{121/121}$, gap $2A/Q$; Appendix C.1). **A30.11/PT sharpened to Case A** — the disputed +1 is the Z_2 -odd seam mode (A28 §12); the ZS-A7 signed seam witness $\hat{u}_{\text{seam}}$ (linear in the Kraus operators, Hadamard-measurable, 4π -closing) makes it a PHYSICAL background observable, not gauge-exact, so the denominator is 121 ($q_{\Lambda} = 83/121$, not $82/120$); DERIVED-CONDITIONAL on $\hat{u}_{\text{seam}} \in H^0(s)$. **A30.12/full** — the 3-form Hamiltonian constraint count ($C(4,p)$ components, $C(2,p)$ physical d.o.f.) confirms 4 non-dynamical components with nonzero F_4 , $w = -1$ (Appendix C.2). **A30.14 entropic-typicality selection** — under a uniform-over-microstates measure $P(x) \propto \exp(N_{\text{eff}} S_{\text{cg}}(x))$, $83/121$ is the MODAL observed Ω_{Λ} (parameter-free peak; curvature $S_{\text{cg}}'' = -121^2/(83 \cdot 38) = -4.642$; spread N_{eff} -dependent), upgrading Escape-2 from existence to selection; DERIVED-CONDITIONAL on the measure (Appendix C.3). **Corrections:** the Branch-A/B dichotomy is credited to ZS-A28 §16.7's PROVEN 'top-form XOR single-trace' dilemma (dynamical form, not re-claimed); and the $\text{Vec}_{\{Z_{11}\}}$ fusion route is **RETRACTED-as-proposed** (§8.6) — B_Z carries $\text{FPdim}(3,2,6)$, $\Sigma d^2 = 49 = 7^2 \neq 11$, so it is not pointed and $\neq \text{Vec}_{\{Z_{11}\}}$; only the $121 = Q^2 = \dim(\text{TV double})$ count survives. **Flagship unchanged:** $\Omega_{\Lambda,0} = 83/121$ remains an unconditional OCCUPATION and a present-epoch ENERGY fraction; with the occupation dynamically complete, PT sharpened, the top-form classified, the No-Go and its escapes classified, and the categorical over-reach excluded, the internal program is COMPLETE — external adversarial review (operator algebra + cosmology) is the next step, and this is the last internally-produced version. Verification: CORE-THEOREM 21, IMPORTED-PROVEN 8, REGRESSION 4, SCOPE-AUDIT 10, all consistent. No new fitted parameters. Consolidated up to v1.5.0. ($A = 35/437$, $Q = 11$, $\dim Z = 2$) LOCKED.

v1.6 (June 2026): Closes the two residuals the external review flagged and repairs the contradictory checks; terminal internal version. **A30.11/PT closed to Case A by a witness functional** — replacing v1.5's 'measurable $\Rightarrow$ physical' non-sequitur, the β -twisted trace $W = \text{STr}_{\beta}$ (β = the ZS-F23 seam Z_2 grading) satisfies $W \circ s = 0$ (a supertrace annihilates graded BRST commutators, verified for even and odd operators) and $W(\hat{u}_{\text{seam}}) \neq 0$ (the ZS-A7 2π -odd seam witness), so $\hat{u}_{\text{seam}} = s\chi$ would give $W(\hat{u}_{\text{seam}}) = 0$, a contradiction — hence $\hat{u}_{\text{seam}}$ is BRST non-exact, the Z_2 -odd seam mode is physical, and the denominator is 121 ($q_{\Lambda} = 83/121$). DERIVED-CONDITIONAL on $\text{BRST-innerness} + \beta = \text{seam grading} + \hat{u}_{\text{seam}} = \text{ZS-A7 representative}$ (Appendix C.4). **A30.14 entropic typicality from exact rank-multinomial counting** — $K_{\Lambda} \sim \text{Binomial}(N_{\text{eff}}, 83/121)$ has EXACT mean $83/121$ and variance $(83 \cdot 38)/(121^2 N_{\text{eff}})$; by Stirling the log-pmf is $N_{\text{eff}} S_{\text{cg}}(x)$, so the $\exp(N_{\text{eff}} S_{\text{cg}})$ measure is the large- N limit of exact counting, with $1/(N_{\text{eff}} S_{\text{cg}}'')$ matching the binomial variance for every N_{eff} . The peak/mean $83/121$ is parameter-free; the width $\propto N_{\text{eff}}^{-1/2}$; the exchangeability axiom is the rank-to-energy map UN , so it relocates rather than solves (Appendix C.3). **A30.10 analytic multiplicity proof** — the coherence eigenvalues $-\frac{1}{2}\gamma(d_i + d_j)$ on the degree-(2,9) mediator graph ($X/Y \leftrightarrow X/Y \rightarrow -2A/Q \times 72$; $X/Y \leftrightarrow Z \rightarrow -A/2 \times 36$; $Z \leftrightarrow Z \rightarrow -9A/Q \times 2$) plus the population spectrum (0, $-2A/Q \times 8$, $-9A/Q$, $-A$) reproduce (1,80,36,3,1) by construction, upgrading C19 from a numeric match to a proof (Appendix C.1). **Repairs:** the broken C14 ($\dim H^0 = 0$ toy that asserted survival) is replaced by the witness gate ($\dim H^0 = 1$, surviving class detected by the cocycle); I6 drops the Z_{11} -pointed claim (FHK handle operator $H = I$ only); the $\text{Vec}_{\{Z_{11}\}} / \text{Dijkgraaf-Witten } Z_{11} \times Z_{11}$ 'translation-transitivity' gloss is RETRACTED in the abstract, §5.3, Appendix B, and the v1.2 history (it conflicted with B_Z carrying $\text{FPdim}(3,2,6)$; equal weight rests on Pachner invariance); the face algebra is written $D_{121} \simeq \mathbb{C}^{121} \subseteq M_{121}$; the 'escape exactly two' wording is narrowed to 'within the declared two-component GR framework'; 'Zero Free Parameters' is restated as 'zero new fitted Z-Spin parameters · structural conditions explicit · one open present-epoch normalization'; and the script docstring/version is unified to v1.6. **Flagship unchanged:** $\Omega_{\Lambda,0} = 83/121$ remains an unconditional OCCUPATION and a present-epoch ENERGY fraction; with PT closed to Case A, the typicality measure derived, the spectrum proved, and the review defects fixed, the internal program is COMPLETE — external adversarial review is the next step. Verification: CORE-THEOREM 23, IMPORTED-PROVEN 8, REGRESSION 4, SCOPE-AUDIT 10, all consistent. Zero new fitted Z-Spin parameters. Consolidated up to v1.6.0. ($A = 35/437$, $Q = 11$, $\dim Z = 2$) LOCKED.